IGEM Tutorial

Version Dev

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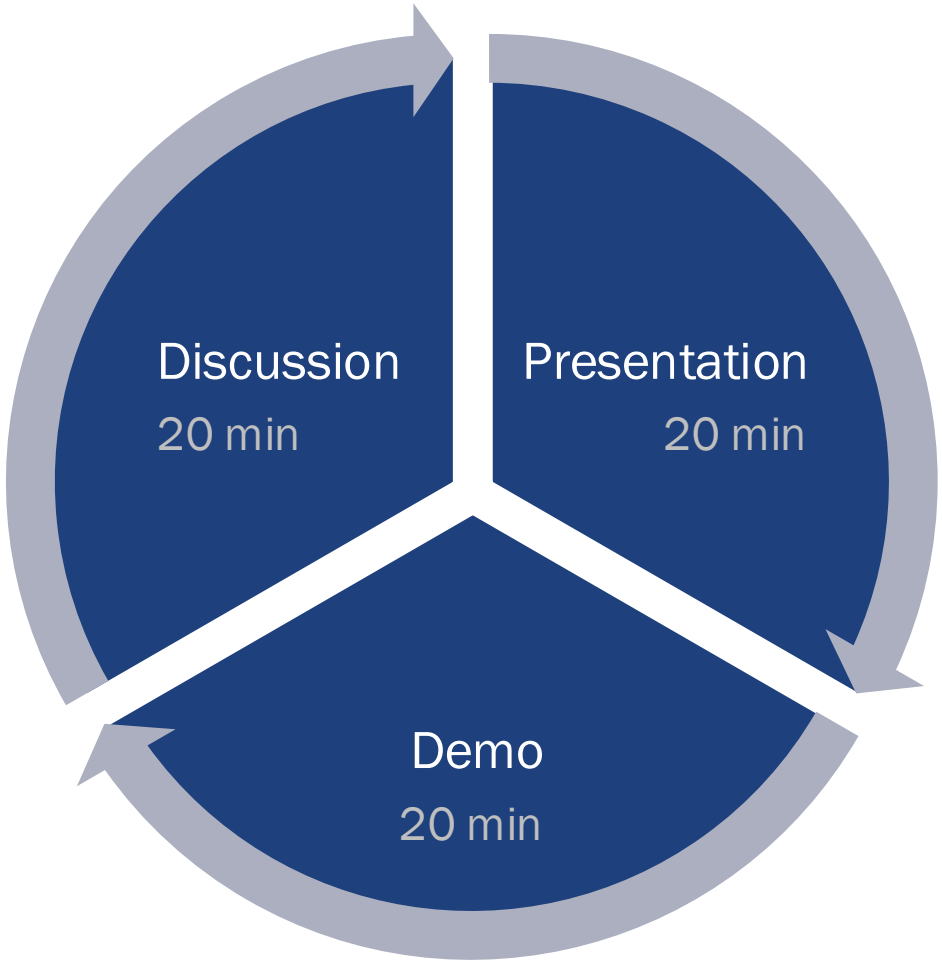
Content



Points we will address:

- About the System
- Initial Settings
- Master Data Customization
- ETL process
- Workflow Monitor
- Data Maintenance
- Data Query
- Use on HPC environment

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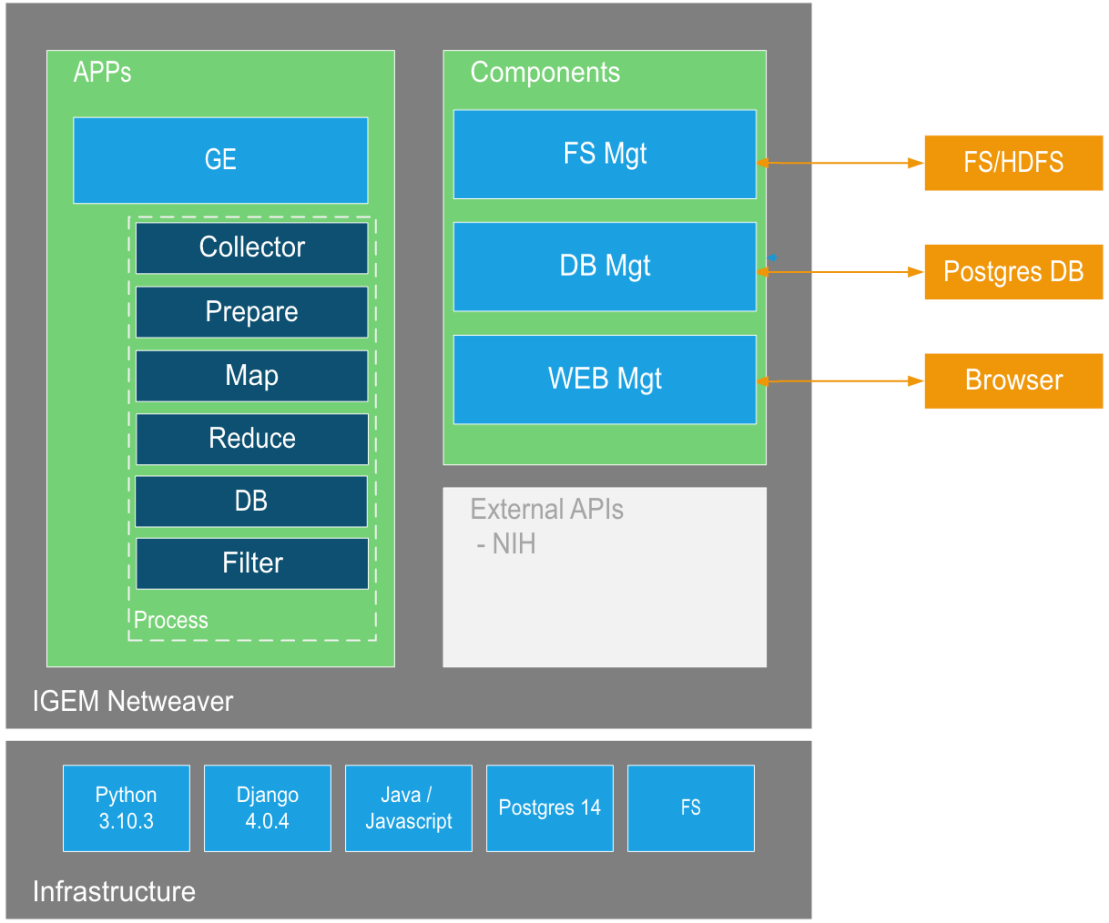
Architecture



The IGEM system was developed as a platform to host different APPs sharing common resources and interacting with each other through a single database

The first APP created was GE (Gene x Exposome) to collect external data sets, identify words, and build a knowledge base available for dynamic and exploratory queries

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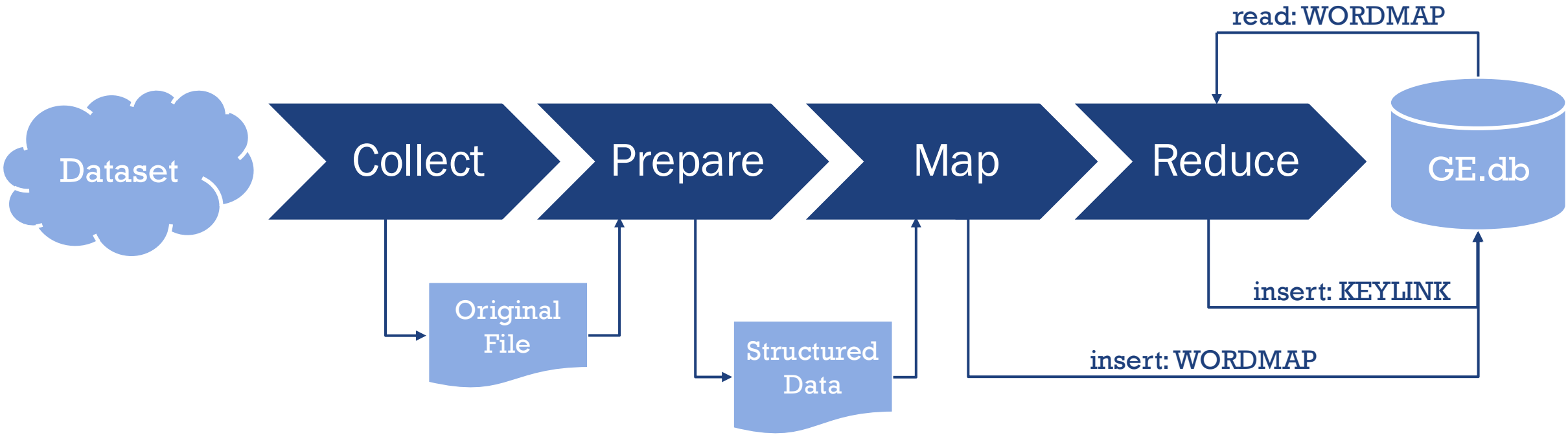
GE.db

The ETL process collects data from external sources, prepare that data to a compatible standard, maps and reduces relationship terms, and writes to GE.db.



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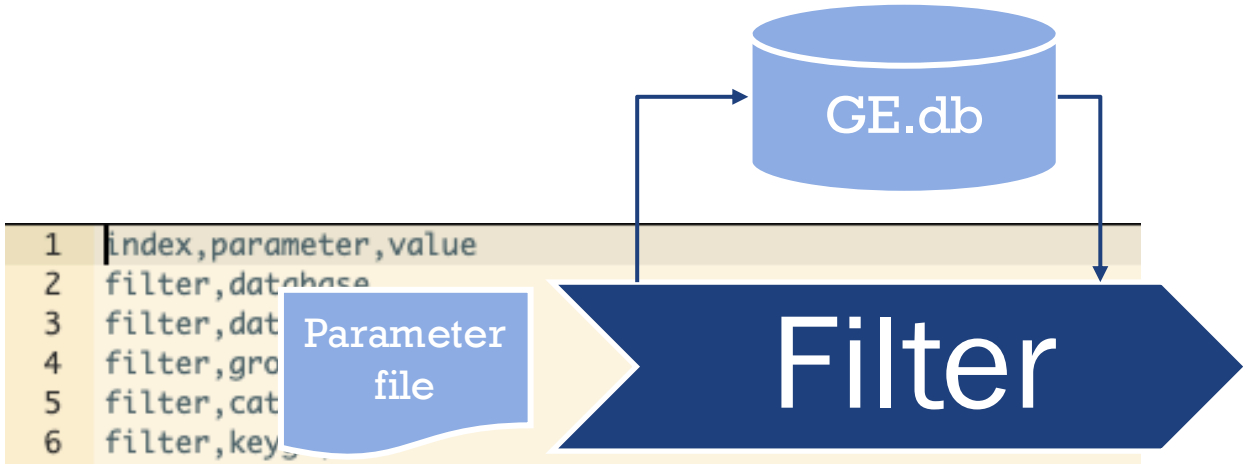
GE.filter

Queries to GE.db in parameter file format




```
index,parameter,value
filter,database
filter,dat
filter,gro
filter,cat
filter,key
filter,word,
output,database,no
output,dataset,no
output,group,
output,category,
output,keyge,no
output,word,no
path,path,/gpfs/group/mah546/default/sw/igem/output.csv
```

Output file

```
database,dataset,group_1,category_1,keyge_1,group_2,category_2,keyge_2,count
,,environment,anatomy,,environment,anatomy,,39318
,,environment,anatomy,,environment,chemical,,103995
,,environment,anatomy,,environment,disease,,10522
,,environment,chemical,,environment,chemical,,133461
,,environment,chemical,,environment,disease,,8367906
,,environment,disease,,environment,disease,,3839
,,environment,anatomy,,genomic,metabolite,,94644
,,environment,anatomy,,genomic,ontology,,12529
t,anatomy,,genomic,pathway,,8378
t,chemical,,genomic,gene,,2418362
t,chemical,,genomic,metabolite,,223413
t,chemical,,genomic,ontology,,6818145
t,chemical,,genomic,pathway,,11492
,,environment,disease,,genomic,gene,,97754796
,,environment,disease,,genomic,metabolite,,908810
,,environment,disease,,genomic,ontology,,2822990
,,environment,disease,,genomic,pathway,,1093
,,genomic,metabolite,,genomic,metabolite,,90828
,,genomic,metabolite,,genomic,pathway,,9172
,,genomic,ontology,,genomic,metabolite,,917246
,,genomic,ontology,,genomic,ontology,,2721
,,genomic,ontology,,genomic,pathway,,836
,,genomic,pathway,,genomic,pathway,,154
```

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HPC – PSU/ICDS Folder

/gpfs/group/mah546/default/sw/igem

- /igem_db/ POSTGRES database structure
- /igem_env/ conda virtual environment files
- /src/ system, app and supports folders.
  - /ge/ codes and interfaces for the functioning of GE APP
  - /loader/ files for loading master data and output directory
  - /psa/ database files downloaded and processed
  - /src/ configurations and parameterizations

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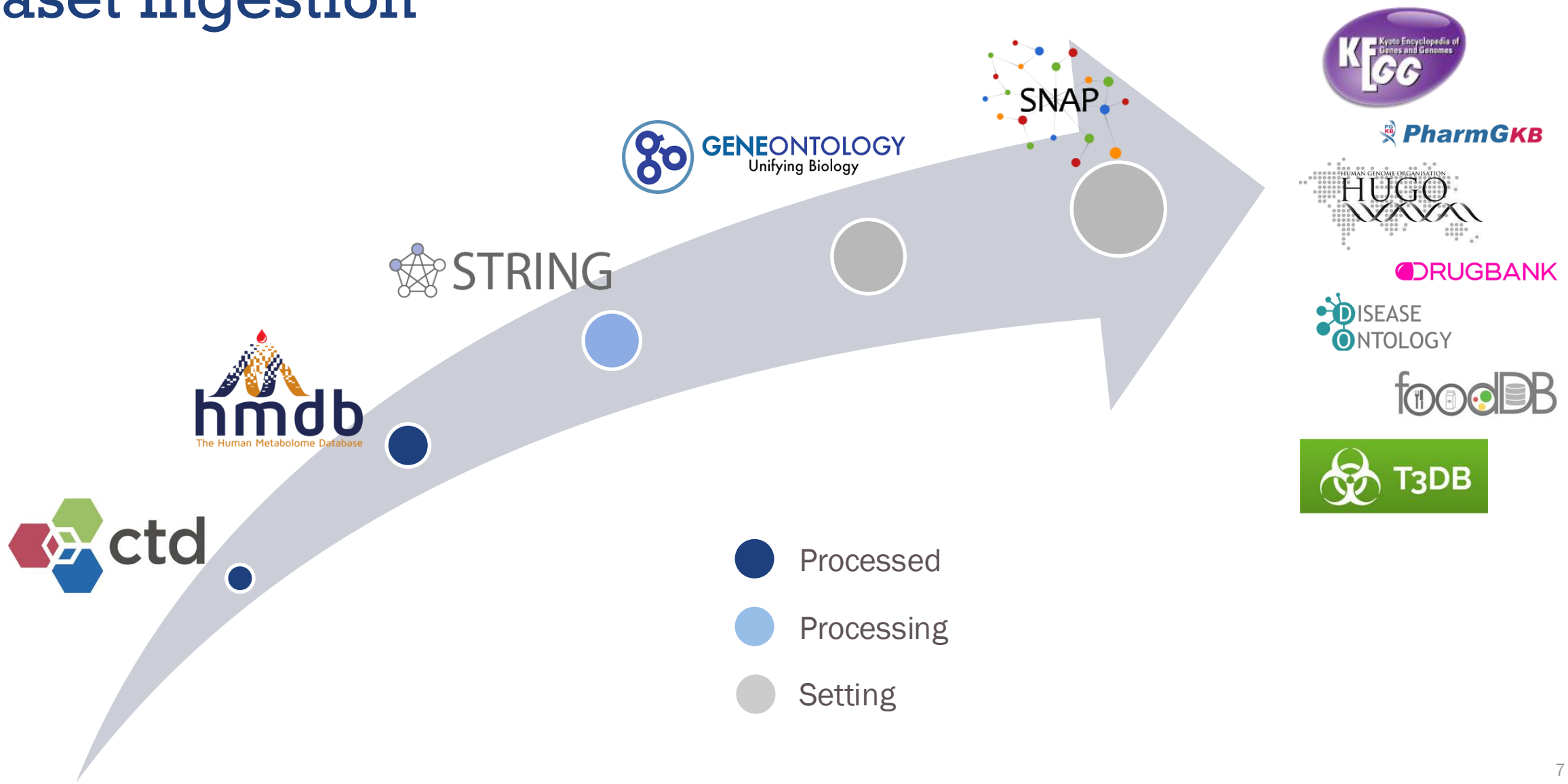
Dataset Ingestion



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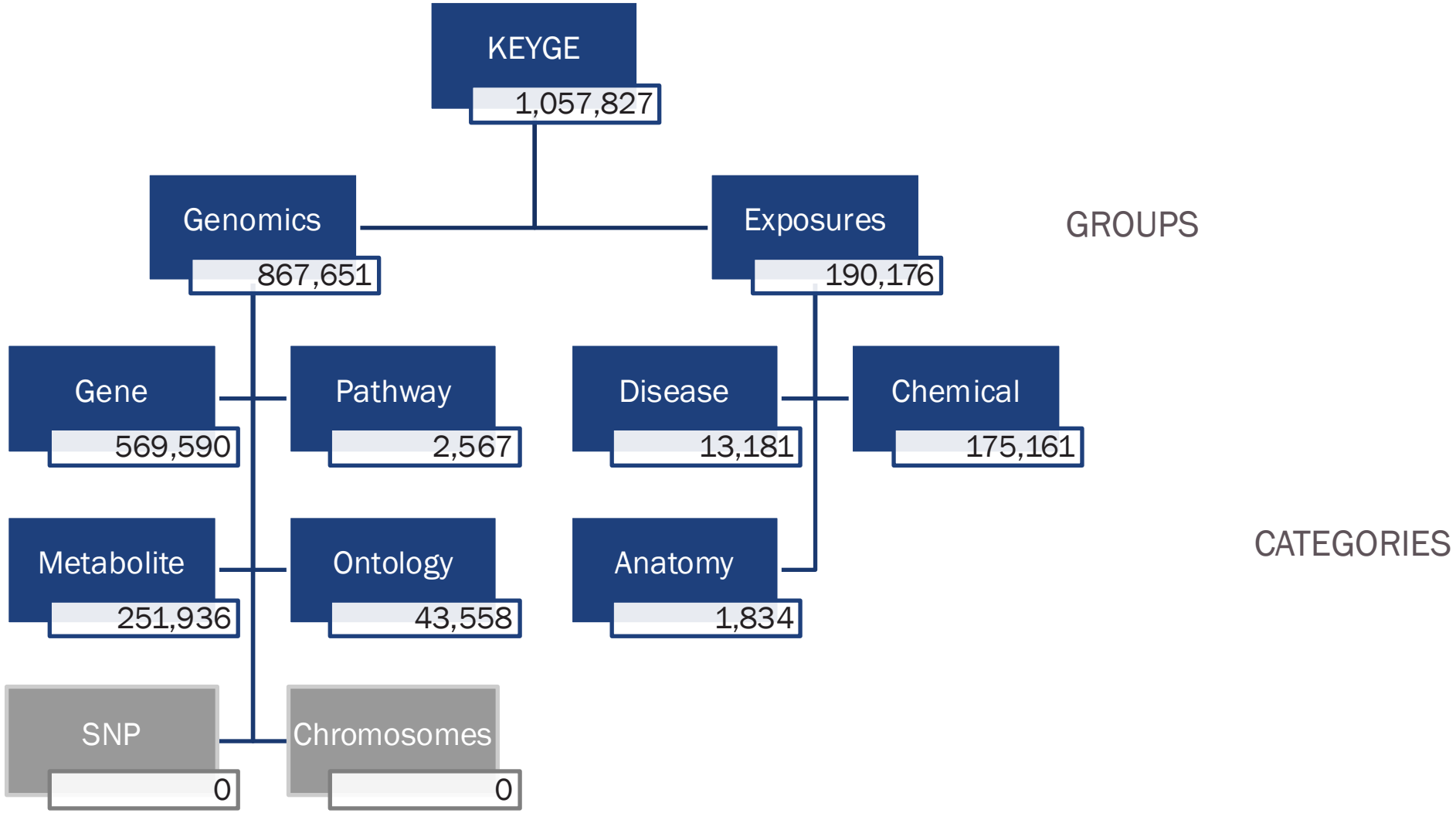
Keyge - Numbers



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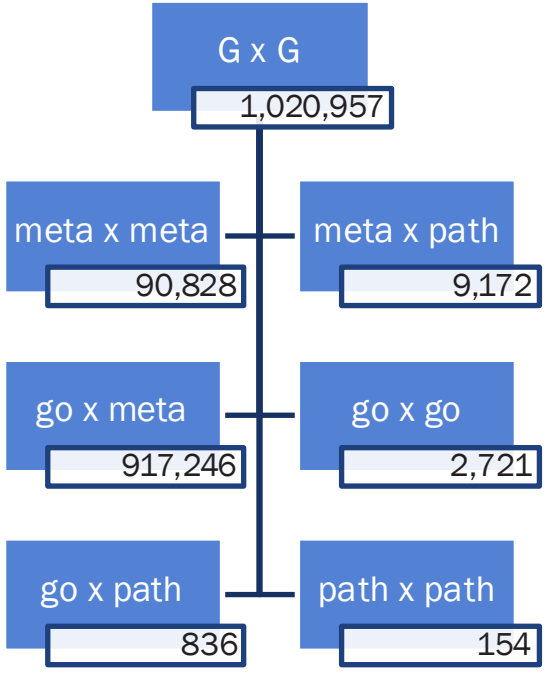
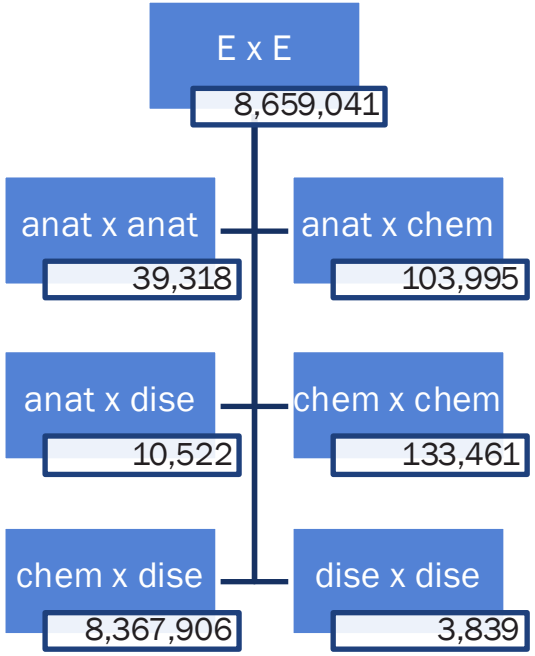
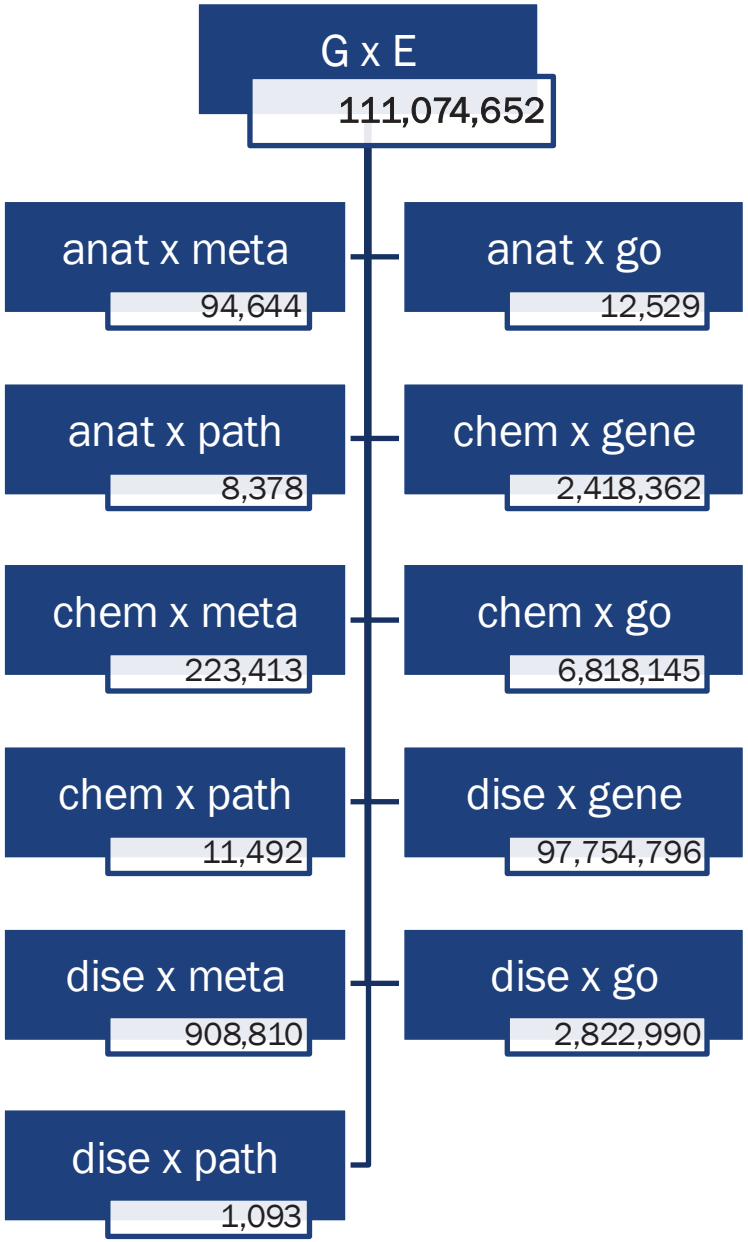
KeyLinks - Numbers

PROCESSED (ctd and hmdb databases)

121 MM links







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Prepare Process

Text from a hypothetical dataset row

lead poisoning, nervous system, childhood silicate niobate lead magnesium niobate. Lead silicate and Lead ion transmembrane transporter activity on childhood lead methacrylate-2-ethylhexanoate spliced leader 30 kda protein, putative ikba variant leads to eda-id lead sulfide oxidation, sulfide, and nitrate.

description character varying (400)	keyge character varying (40)
lead ion transmembrane transporter activity	go:0015094
lead ion transport	go:0015692
lead	meta:hmdb0004628
triphenyllead acetate	chem:c002019
cellular response to lead ion	go:0071284
lead sulfide oxidation	go:0019327
response to lead ion	go:0010288
lead ion binding	go:0032791
lead acetate	chem:c008261
trilead tetroxide formalin	chem:c003696
lead acetate, basic	chem:c008262
lead nitrate	chem:c017461
lead tetrachloride	chem:c017462
lead tetroxide	chem:c028356
lead haematoxylin	chem:c018328
lead sulfide	chem:c018391
lead phosphate	chem:c036682
lead tetraacetate	chem:c029433
lead chromate	chem:c031895
lead chloride	chem:c029891
lead bromide	chem:c032721
lead sulfate	chem:c032722
triethyllead acetate	chem:c033120
lead silicate	chem:c044795
lead carbonate	chem:c043262
lead azide	chem:c037372

How many *meta:hmdb0004628* does the text have?

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Prepare Process

Text from a hypothetical dataset row

lead poisoning, nervous system, childhood silicate niobate lead magnesium niobate. Lead silicate and Lead ion transmembrane transporter activity on childhood lead methacrylate-2-ethylhexanoate spliced leader 30 kda protein, putative ikba variant leads to eda-id lead sulfide oxidation, sulfide, and nitrate.

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lead tetroxide	chem:c028356
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lead sulfide	chem:c018391
lead phosphate	chem:c036682
lead tetraacetate	chem:c029433
lead chromate	chem:c031895
lead chloride	chem:c029891
lead bromide	chem:c032721
lead sulfate	chem:c032722
triethyllead acetate	chem:c033120
lead silicate	chem:c044795
lead carbonate	chem:c043262
lead azide	chem:c037372

How many *meta:hmdb0004628* does the text have?

Answer: 0 (zero)

All keywords “lead” in the text are compounded with other words

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Prepare Process

ZIP Sweet metabolites.xml

hmdb
The Human Metabolome Database

Collect → Prepare

description

meta:hmdb0000008 meta:hmdb0000039 meta:hmdb0000125 chem:d005947 meta:hmdb0000742 chem:d013912 chem:d008715 dise:d001237 chem:d003545 dise:d007333 chem:d003540 go:0019346 meta:hmdb0001275 go:0034440 dise:d056693 meta:hmdb0001967 path:r-hsa-211981 go:0098754 path:r-hsa-1430728 anat:d014024 meta:hmdb0000661 chem:d019344 meta:hmdb0000574 meta:hmdb0000167 dise:d000138 anat:d002490 dise:d008659 anat:d014556 anat:d010949 go:0007588 anat:d008099

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	AA	AB	AC	AD	AE	AF	AG	AH	AI	AJ	AK	AL	AM	AN	AO	AP	AQ	AR	AS
1	version	creatior	update_	accessi	status	seconda	name	description	synonyi	chemic	average	monisc	iupac_r	traditic	cas_reg	smiles	inchi	inchike	taxonoi	ontolog	state	experin	predict	spectra	biologi	normal	abnorm	disease	foodb_	kegg_ic	chemsp	chebi_i	pubche	pdb_id	biocyc_	drugba	phenol_	wikipec	knapsa	bigg_id	metlin_	vmh_id	synthes	genera	protein	fb
2	5	2005-1	2021-0	HMDB0	quantified		2-Hydro	2-Hydroxybutyric aci		C4H8O	104.1	104	(2S)-2-	(S)-2-h	3347-9	CC[C@	InChI=1	AFENDNXGAFYKQO-V	Solid									FDB02	C05984	4E+05	50613	4E+05		CPD-3564			2-Hydroxybutyric_acid					Carlier, J. P.; Henry, C.; Lo	
3	5	2005-1	2021-0	HMDB0	quantified		Creatin	Creatine, is a natura		C4H9N	131.1																						86	CREATI	DB00148		Creatine		34543	7	CREAT	Thalhammer, Franz; Gast	
4	5	2005-1	2021-1	HMDB0	quantified		cis-Aco	cis-Aconitic acid is a		C6H6O	174.1																						05	CIS-ACONITATE			Citric a	C00001177		5130	HC0034	Gutierrez, Eddie N.; Lamb	
5	5	2005-1	2020-1	HMDB0	quantified		Dimeth	Dimethylamine (DM		C2H7N	45.08																						74	DIMETHYLAMINE			Dimethylamine			3758	DMA	Zones, Stacey I.; Burton,	
6	5	2005-1	2021-0	HMDB0	quantified		Citric a	Citric acid (citrate) is		C6H8O	192.1																						11	CIT	DB04272		Citric_	C00007	34080	124	CIT	Evans, D. W. S. Hydrolysis	
7	5	2005-1	2020-1	HMDB0	quantified		Choline	Choline is a basic co		C5H14N	104.2																						05	CHOLIN	DB00122		Choline	C00007	33910				
8	5	2005-1	2020-1	HMDB0	quantified		Ethanol	Ethanol is a clear, co		C2H6O	46.07																						02	ETOH	DB00898		Ethano	C00019	35062				
9	5	2005-1	2020-1	HMDB0	quantified		Glycolic	Glycolic acid (or hydi		C2H4O	76.05																						67	GLYCOI	DB03085		Glycoli	C00007	34090				
10	5	2005-1	2021-0	HMDB0	quantified		D-Gluc	Glucose is a monosa		C6H12	180.2																						89	GLC	DB02379		Glucos	C00001122					

description

2-Hydroxybutyric acid (CAS: 600-15-7), also known as alpha-hydroxybutyrate, is an organic acid derived from alpha-ketobutyrate. alpha-Ketobutyrate is produced by amino acid catabolism (threonine and methionine) and glutathione anabolism (cysteine formation pathway) and is metabolized into propionyl-CoA and carbon dioxide (PMID: 20526369). 2-Hydroxybutyric acid is formed as a byproduct from the formation of alpha-ketobutyrate via a reaction catalyzed by lactate dehydrogenase (LDH) or alpha-hydroxybutyrate dehydrogenase (alphaHBDH). alpha-Hydroxybutyric acid is primarily produced in mammalian hepatic tissues that catabolize L-threonine or synthesize glutathione. Oxidative stress or detoxification of xenobiotics in the liver can dramatically increase the rate of hepatic glutathione synthesis. Under such metabolic stress conditions, supplies of L-cysteine for glutathione synthesis become limiting, so homocysteine is diverted from the transmethylation pathway (which forms methionine) into the transsulfuration pathway (which forms cystathionine). alpha-Ketobutyrate is released as a byproduct when cystathionine is cleaved into cysteine that is incorporated into glutathione. Chronic shifts in the rate of glutathione synthesis may be reflected by urinary excretion of 2-hydroxybutyrate. 2-Hydroxybutyrate is an early marker for both insulin resistance and impaired glucose regulation that appears to arise due to increased lipid oxidation and oxidative stress (PMID: 20526369). 2-Hydroxybutyric acid is often found in the urine of patients suffering from lactic acidosis and ketoacidosis. 2-Hydroxybutyric acid generally appears at high concentrations in situations related to deficient energy metabolism (e.g. birth asphyxia) and also in inherited metabolic diseases affecting the central nervous system during neonatal development, such as "cerebral" lactic acidosis, glutaric aciduria type II, dihydrolipoyl dehydrogenase (E3) deficiency, and propionic acidemia. More recently it has been noted that elevated levels of alpha-hydroxybutyrate in the plasma is a good marker for early-stage type II diabetes (PMID: 19166731). It was concluded from studies done in the mid-1970s that an increased NADH2/NAD ratio was the most important factor for the production of 2-hydroxybutyric acid (PMID: 168632).

Hmdbsweet.csv

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Prepare Process

keyge character varying (40)	description character varying (400)
chem:c008861	phenyl-gamma-methylmercapto-alpha-hydroxybutyrate
chem:c027745	alpha-aceto-alpha-hydroxybutyrate

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meta:hmdb0000008	2-Hydroxybutyric acid	metabolite
meta:hmdb0000125	glutathione	metabolite
meta:hmdb0000742	homocysteine	metabolite
meta:hmdb0001275	propionyl-coa	metabolite
meta:hmdb0001967	carbon dioxide	metabolite
meta:hmdb0000661	glutaric aciduria	metabolite
meta:hmdb0000167	l-threonine	metabolite
meta:hmdb0000574	l-cysteine	metabolite
chem:d005947	glucose	chemical
chem:d013912	threonine	chemical
chem:d008715	methionine	chemical
chem:d003545	cysteine	chemical
chem:d003540	cystathionine	chemical
chem:d019344	lactic acid	chemical
dise:d001237	asphyxia	disease
dise:d007333	insulin resistance	disease
dise:d000138	acidosis	disease
dise:d056693	propionic acidemia	disease
dise:d008659	metabolic diseases	disease
go:0019346	transsulfuration	ontology
go:0034440	lipid oxidation	ontology
go:0098754	detoxification	ontology
go:0007588	excretion	ontology
path:r-hsa-211981	xenobiotics	pathway
path:r-hsa-1430728	metabolism	pathway
anat:d014024	tissues	anatomy
anat:d002490	central nervous system	anatomy
anat:d014556	urine	anatomy
anat:d010949	plasma	anatomy
anat:d008099	liver	anatomy
gene:31251940	dihydrolipoyl dehydrogenase	gene

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Critical Point

Cases where the same word refers to two different Keyge.

description character varying (400)	keyge character varying (40)
lead	meta:hmdb0004628
lead	chem:d007854

The current version will depend on the analysis at the time of querying the data. For the following versions, we can think of intelligence to differentiate the context of the word and suggest the best option.

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BACKUPS

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Critical Point

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meta:hmdb0000008	- 2-Hydroxybutyric acid
meta:hmdb0000125	- glutathione
chem:d005947	- glucose
meta:hmdb0000742	- homocysteine
chem:d013912	- threonine
chem:d008715	- methionine
dise:d001237	- asphyxia
chem:d003545	- cysteine
dise:d007333	- insulin resistance
chem:d003540	- cystathionine
go:0019346	- transsulfuration
meta:hmdb0001275	- propionyl-coa
go:0034440	- lipid oxidation
dise:d056693	- propionic acidemia
meta:hmdb0001967	- carbon dioxide
path:r-hsa-211981	- xenobiotics
go:0098754	- detoxification
path:r-hsa-1430728	- metabolism
anat:d014024	- tissues
meta:hmdb0000661	- glutaric aciduria
chem:d019344	- lactic acid
meta:hmdb0000574	- L-cysteine
meta:hmdb0000167	- l-threonine
dise:d000138	- acidosis
anat:d002490	- central nervous system
dise:d008659	- metabolic diseases
anat:d014556	- urine
anat:d010949	- plasma
go:0007588	- excretion
anat:d008099	- liver

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Content

- About the System
- Initial Settings
- Master Data Customization
- ETL process
- Monitor the workflow
- Data Maintenance
- Data Query
- Raise the process in the HPC environment

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Initial Settings

About the System

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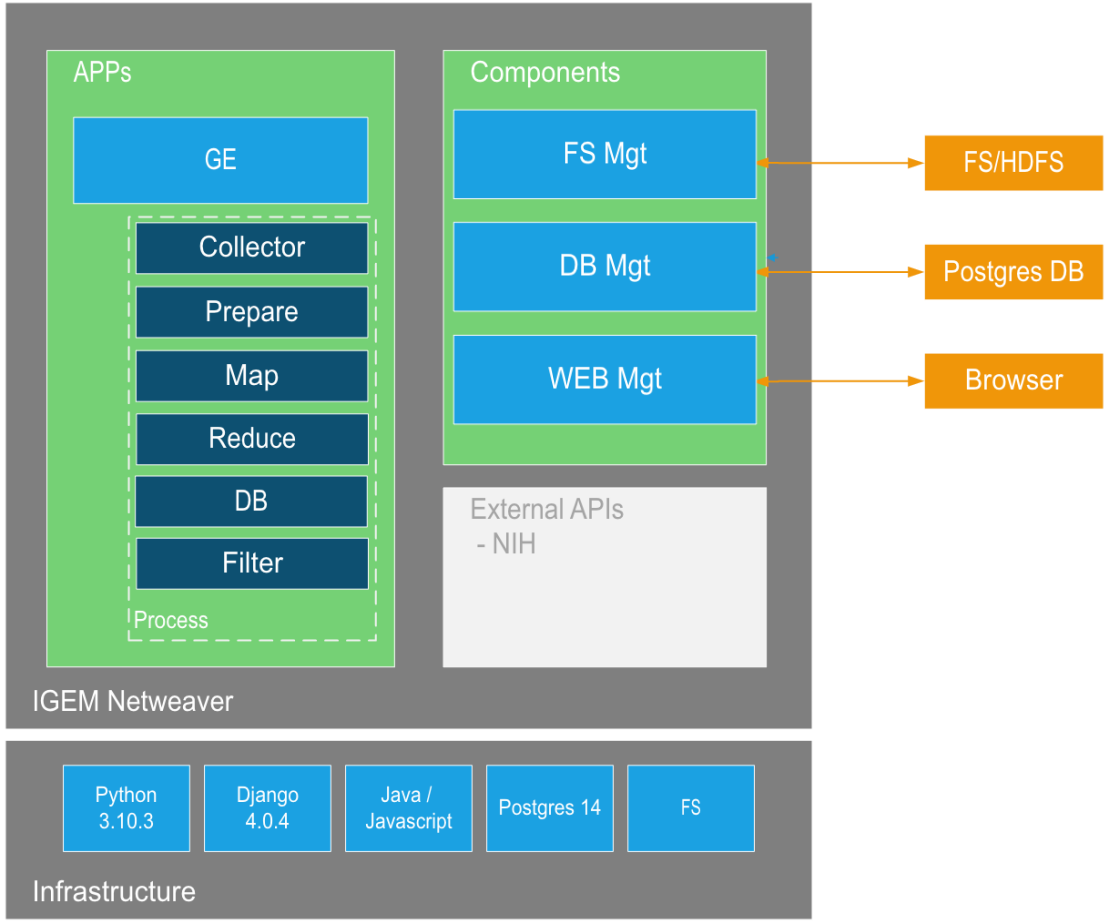
Architecture



The IGEM system was developed as a platform to host different APPs sharing common resources and interacting with each other through a single database

The first APP created was GE (Gene x Exposome) to collect external data sets, identify words, and build a knowledge base available for dynamic and exploratory queries

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GE APP Process

Data Workflow

- collector
- prepare
- map
- Reduce

Filter and Queries

- filter

DB

- load
- download
- show
- truncate
- delete

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HPC – PSU/ICDS Folder

/gpfs/group/mah546/default/sw/igem

- /igem_db/ POSTGRES database structure
- /igem_env/ conda virtual environment files
- /src/ system, app and supports folders.
  - /ge/ codes and interfaces for the functioning of GE APP
  - /loader/ files for loading master data and output directory
  - /psa/ database files downloaded and processed
  - /src/ configurations and parameterizations

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HPC – PSU/ICDS Database

/gpfs/group/mah546/default/sw/igem/src/src/settings.py

```
DATABASES = {
    'default': {
        'ENGINE': 'django.db.backends.postgresql_psycopg2',
        'NAME': 'GE',
        'USER': 'postgres',
        'PASSWORD': 'admin2022',
        'HOST': '127.0.0.1',
        'PORT': '5432',
    }
}
```

Database settings linked to POSTGRESql

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Master Datas

Master Data Customization

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Master Data List

Master data will configure in the following order:

1. DATABASE
2. KEYGE-PREFIX
3. DATASET
4. KEYGE-GROUP
5. KEYGE-CATEGORY
6. KEYGE
7. KEYGE-WORD

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Master Data via WEB interface

IGEM admin

WELCOME, **ANDRE**. VIEW SITE / CHANGE PASSWORD / LOG OUT

Home › Ge › Database

Start typing to filter...

AUTHENTICATION AND AUTHORIZATION

Groups	+ Add
Users	+ Add

GE

Database	+ Add
Dataset	+ Add
Dataset - Columns	+ Add
Dataset - Workflow	+ Add
Keyge	+ Add
Keyge - Category	+ Add
Keyge - Group	+ Add
Keyge - Prefix	+ Add
Keyge - Word	+ Add
Links - Keyge	+ Add
Links - Word	+ Add

Add new record

Select database to change

Add new record

ADD DATABASE +

Search

Action: --------- Go 0 of 52 selected

DATABASE	CATEGORY	DESCRIPTION
BacMap	Genomics	BacMap
BovineDB	Genomics	Bovine Metabolome Database
YMDB	Genomics	Yeast Metabolome Database
DataMed	Genomics	Omics Discovery Index
MeSH	Genomics	Medical Subject Headings
ChEBI	Genomics	Chemical Entities of Biological Matter
ChemSpider	Genomics	ChemSpider
PubChem	Genomics	PubChem
Chemnetbase	Genomics	Dictionary of Natural Products
EBI	Genomics	MetaboLights
Metabolomics	Genomics	NIH Metabolomics Workbench

FILTER

By category

All

Genomics

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Master Data via WEB interface

IGEM admin

WELCOME, **ANDRE**. VIEW SITE / CHANGE PASSWORD / LOG OUT

Home › Ge › Database › CTD

Change database

Master Data Fields

HISTORY

CTD

Database:	CTD
Description:	Comparative Toxicogenomics Database
Category:	Genomics
Website:	http://ctdbase.org/

Delete | Save and add another | Save and continue editing | SAVE

Function buttons

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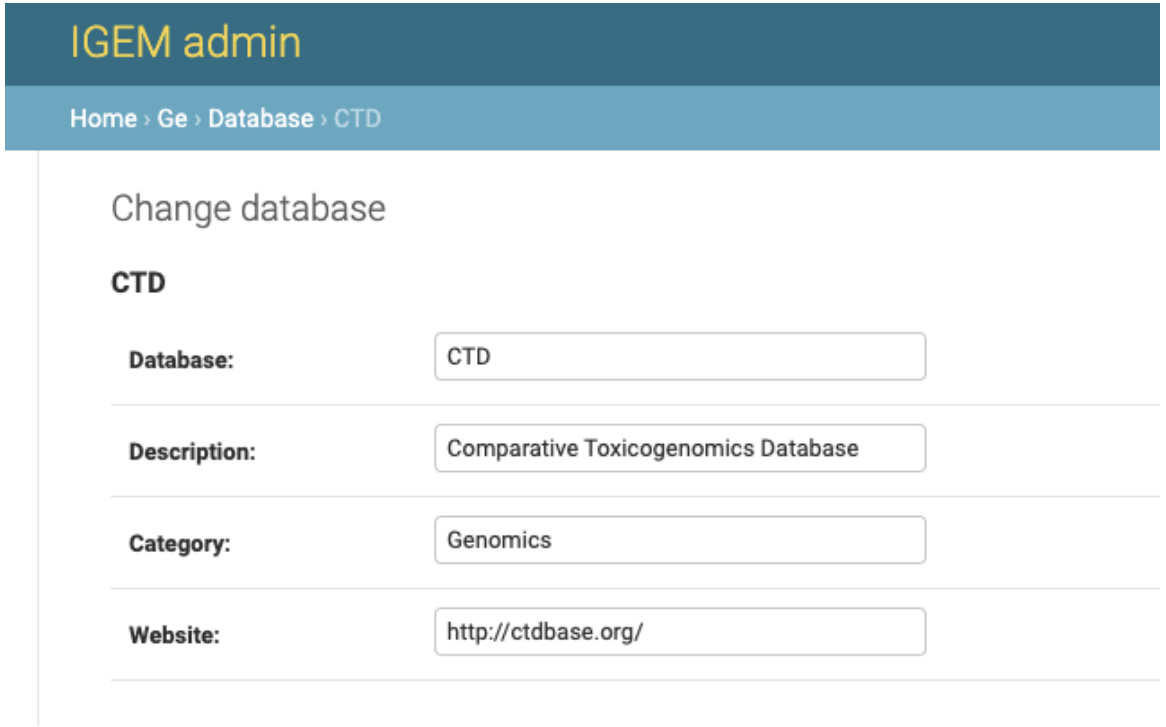
Databases



A Database record points to a specific external dataset, having authentication information, hosted data category, and description. It will reference in the Dataset (subsequent master data).

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Datasets

A Dataset record stores all information needed for external extraction and controls the links of related terms by the mapping process. The control of external data extraction occurs by Dataset

IGEM admin

WELCOME, ANDRE. VIEW SITE / CHANGE PASSWORD / LOG OUT

Home › Ge › Dataset › STRING_CLUSTER

Change dataset

HISTORY

STRING_CLUSTER

Database: STRING

Dataset: STRING_CLUSTER

Description: String

☑ Activate

Attributes (Hide)

☑ Source path from Internet

Source path: https://stringdb-static.org/download/clusters

Source file name: 9606.clusters.info.v11.5.txt.gz

Source file format: GZ

Source file sep: \t

Source file skiprow: 1

☑ Source compact

Target file name: 9606.clusters.info.v11.5.txt

Target file format: TXT

☐ Keep file

DATASET - COLUMNS

COLUMN SEQUENCE	COLUMN NAME	ACTIVE?	PREFIX	DELETE?
DSTColumn object (16) 0	#ncbi_taxid	☑	gene:	☐
DSTColumn object (17) 1	cluster_id	☑	disa:	☐
DSTColumn object (18) 2	cluster_size	☑	chem:	☐

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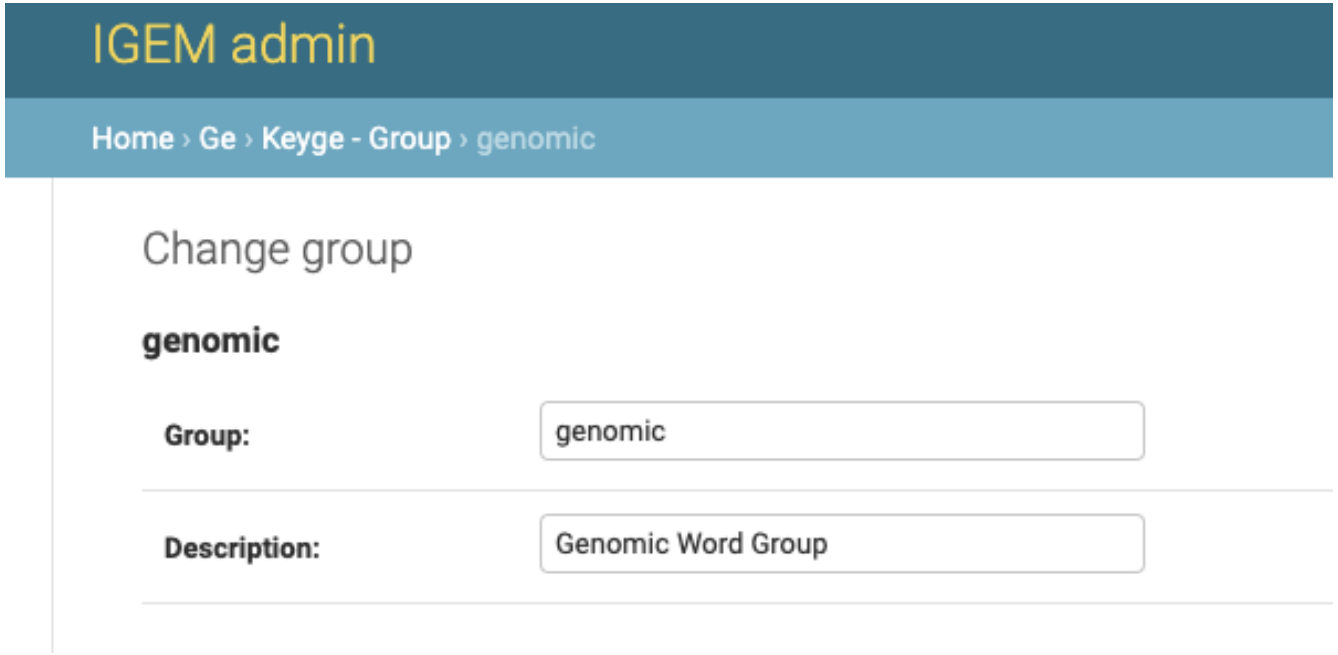
KEYGE - GROUP



The system groups will help in the type of search terms and will be assigned in the KEYGE explained later.

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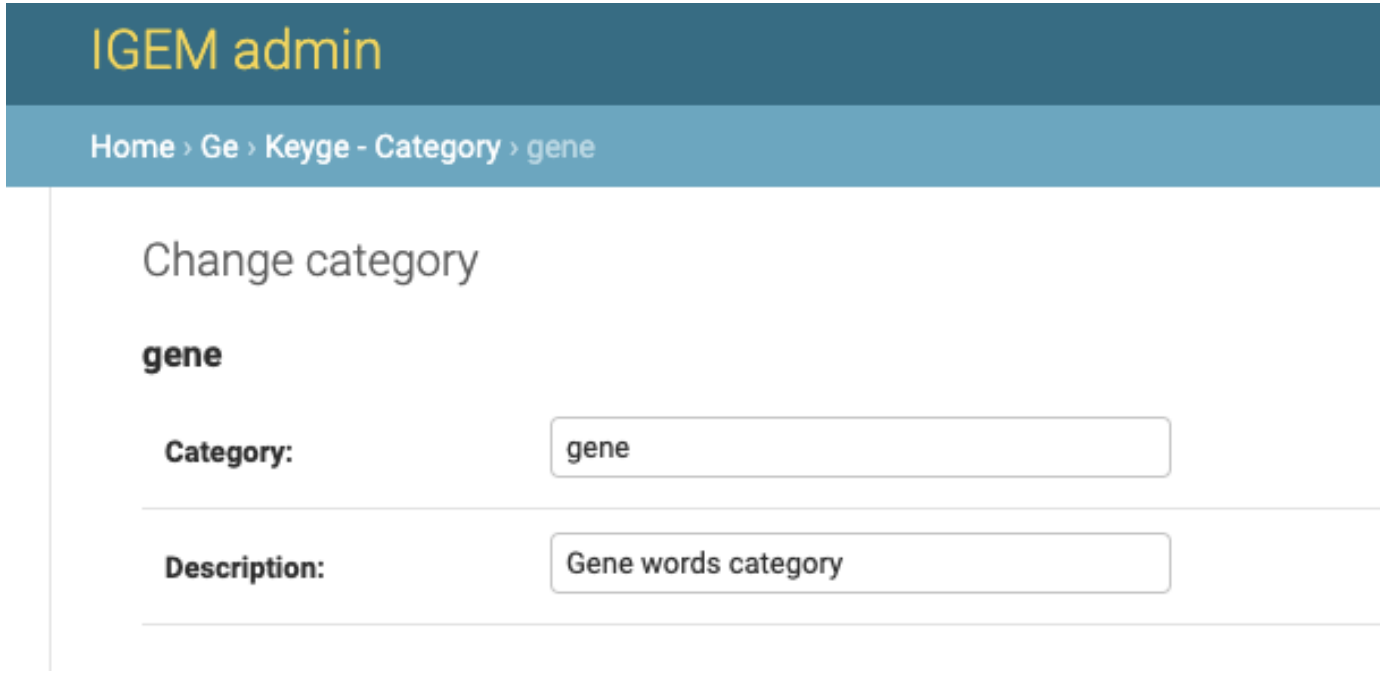
KEYGE - Category



The categories help identify and search the terms and will be assigned in the KEYGE explained in the sequence

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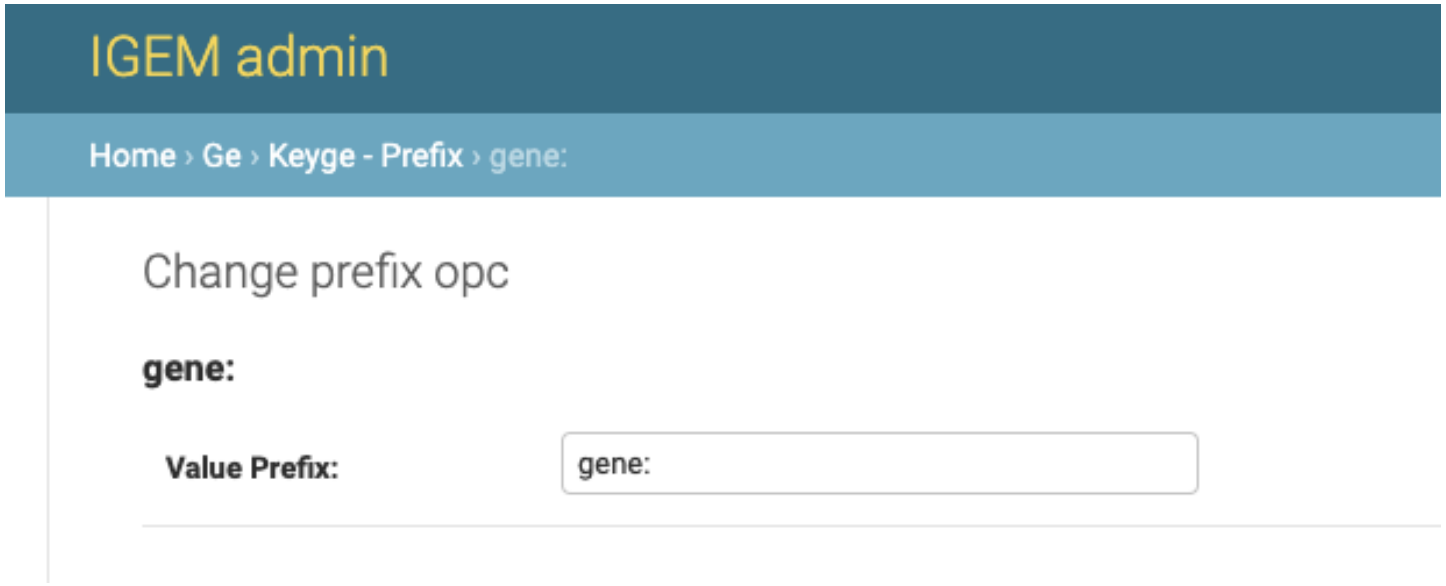
KEYGE - Prefix



Groups to identify the type of information per column of extracted data. This information will be used when parameterizing the Dataset in the Columns group

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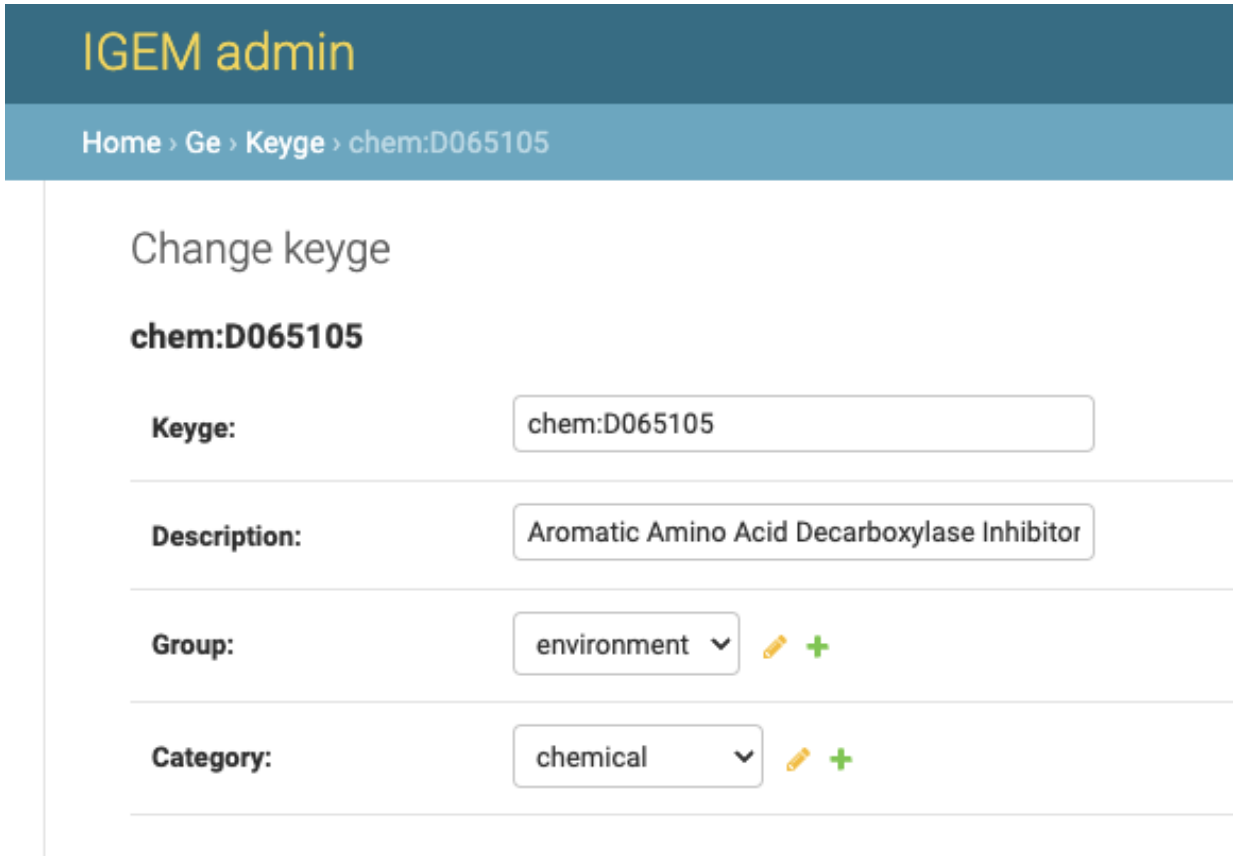
KEYGE



This master data will keep words that will be searched or converted in the ETL process of each DATASET.
The conversion will take place with the support of the KEYGE-WORD register, explained below.
The KEYGE will be the central point at the time of the searches, and the result of the ETL will be the connection of two KEYS.

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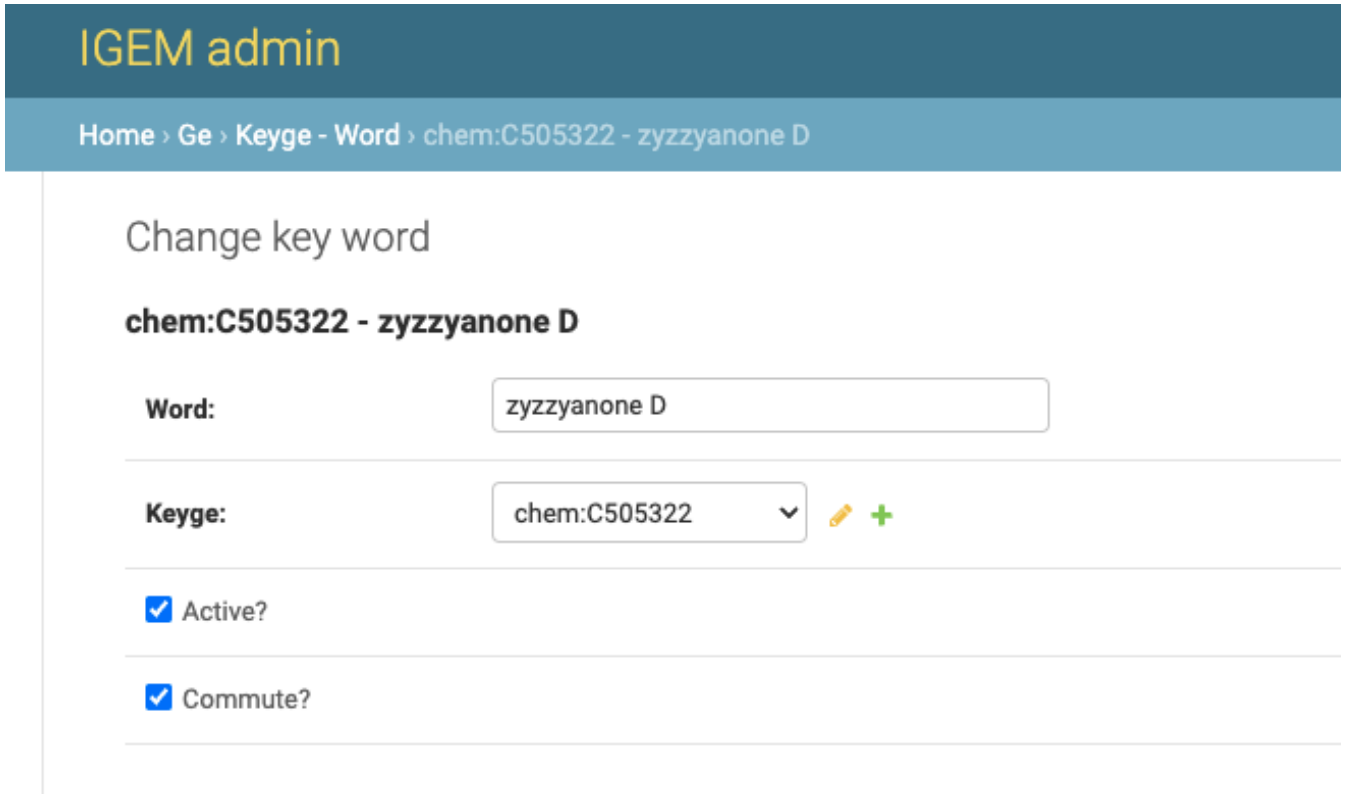
KEYWORD



Master data that will keep the relationship of words and KEYGE. The system allows you to register compound words.

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Command Line

It will be possible to add master records via the command line for batch updates.

```
$ python manage.py db load –database –path .../database.csv
$ python manage.py db load –dataset –path .../dataset.csv
$ python manage.py db load –group –path .../group.csv
$ python manage.py db load –category –path .../category.csv
$ python manage.py db load –keyge –path .../keyge.csv
$ python manage.py db load –keyword –path .../keyword.csv
```

Each Master Data has its format. In the loader folder, there are loading models.

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ETL Processes

Extract, Transformation and Load

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Workflow

The ETL process (Extraction, Transformation, and Loader) will be the step in which the system will fetch data from external sources, treat this data to a compatible standard, search for term relationships and write to GE.db.

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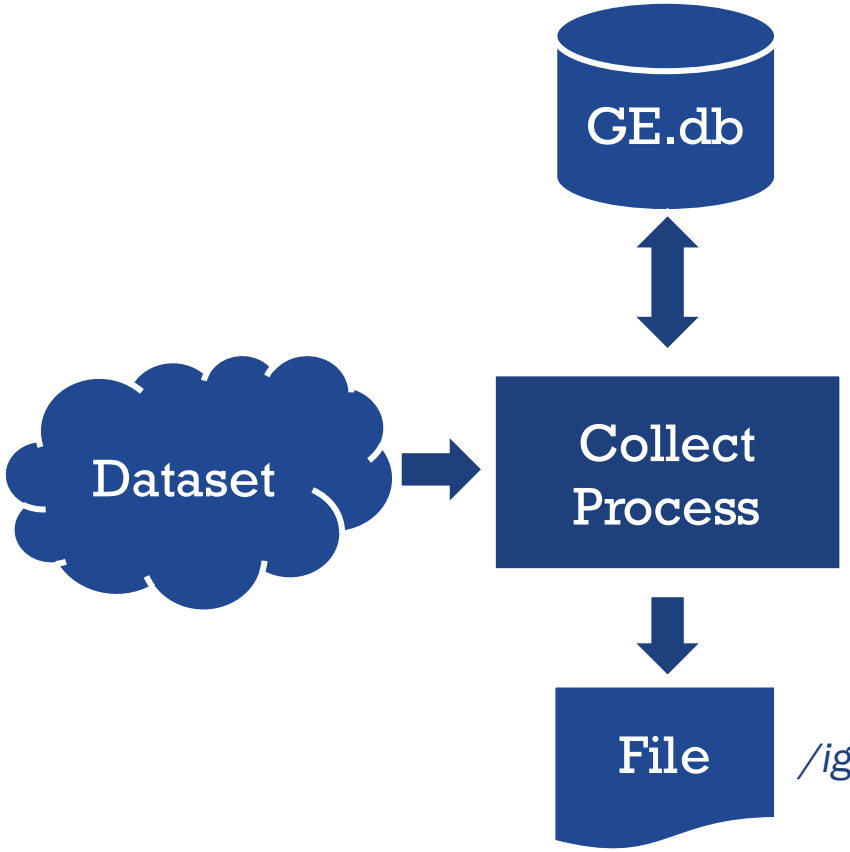
Phase 1 – Collect Process



/igem/src/psa/{dataset}/{file_name}

Command Line:

$ python manage.py collect --run {all or dataset}
$ python manage.py collect --reset {all or dataset}
$ python manage.py collect –show
$ python manage.py collect --activate {dataset}
$ python manage.py collect --deactivate {dataset}

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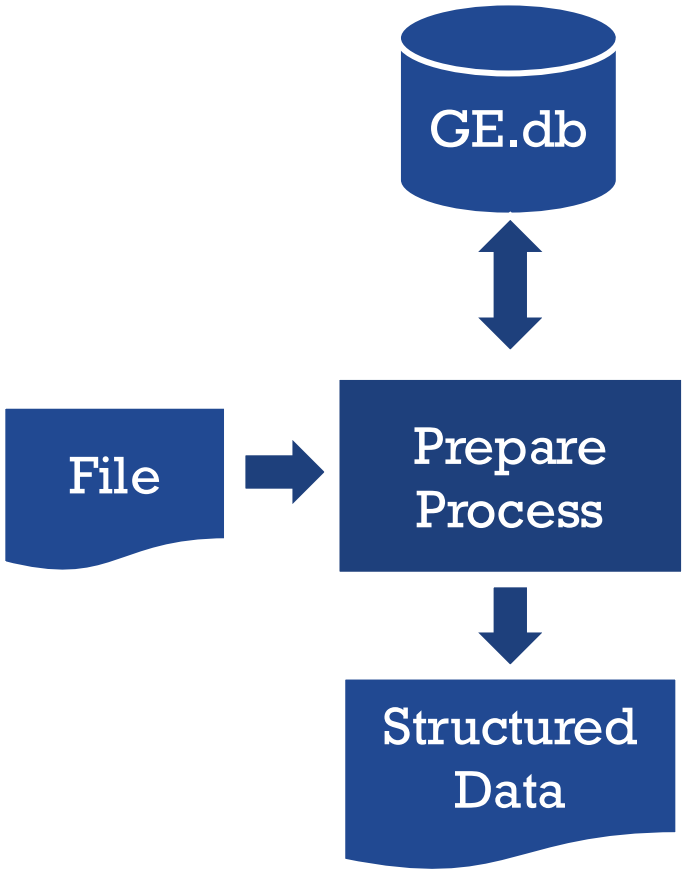
Phase 2 – Prepare Process



- Deleting header lines
- Deleting unnecessary columns
- Transforming ID Columns with Suffix Identifiers
- Replacement the terms
- Deletion of the original file
- Link words to keyge

/igem/src/psa/{dataset}/data.csv

Command Line:

$ python manage.py prepare --run {all or dataset}
$ python manage.py prepare --reset {all or dataset}

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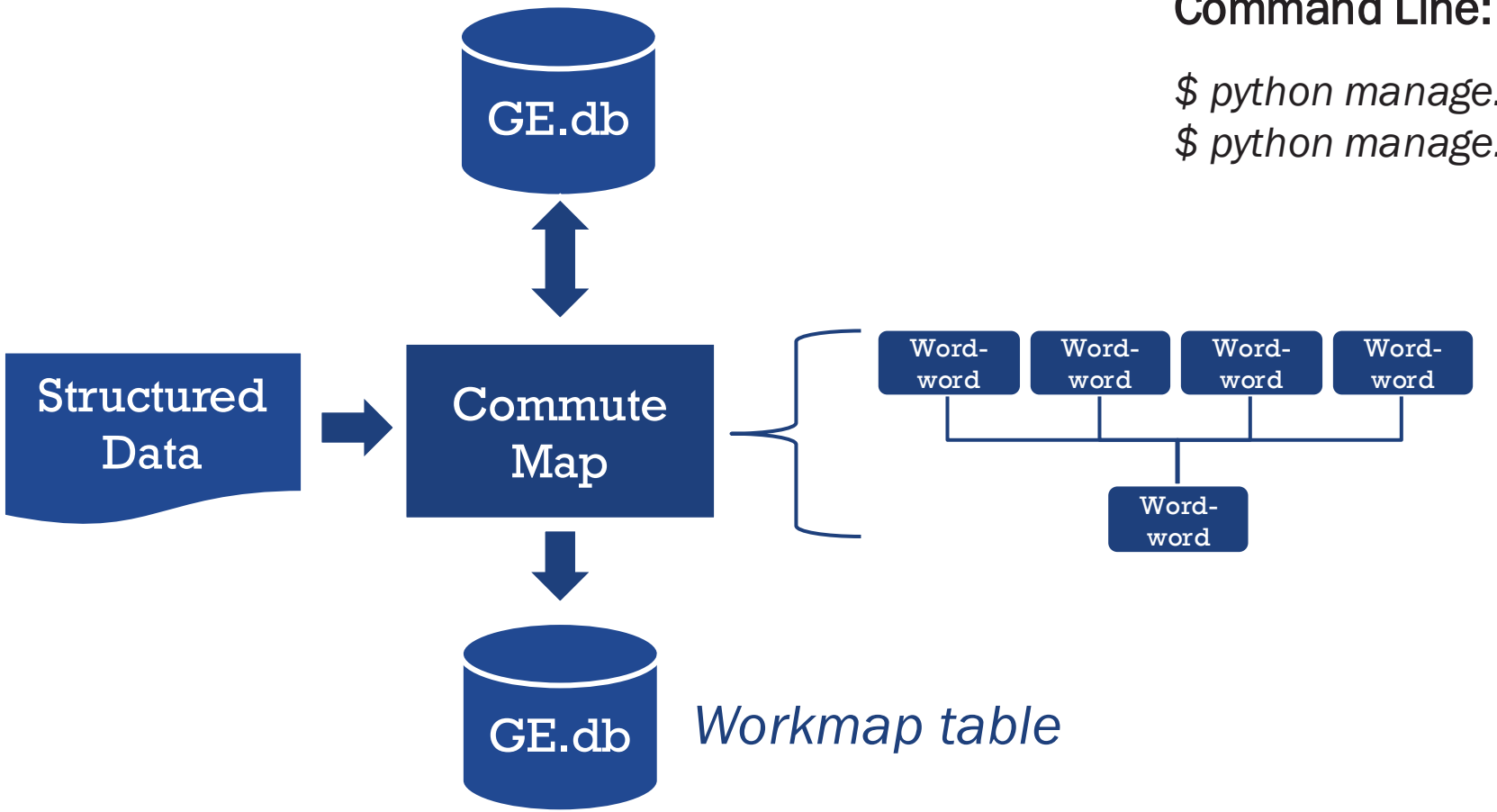
Phase 3 – Map Process



Command Line:

$ python manage.py map –run {all or dataset}
$ python manage.py map –reset {all or dataset}

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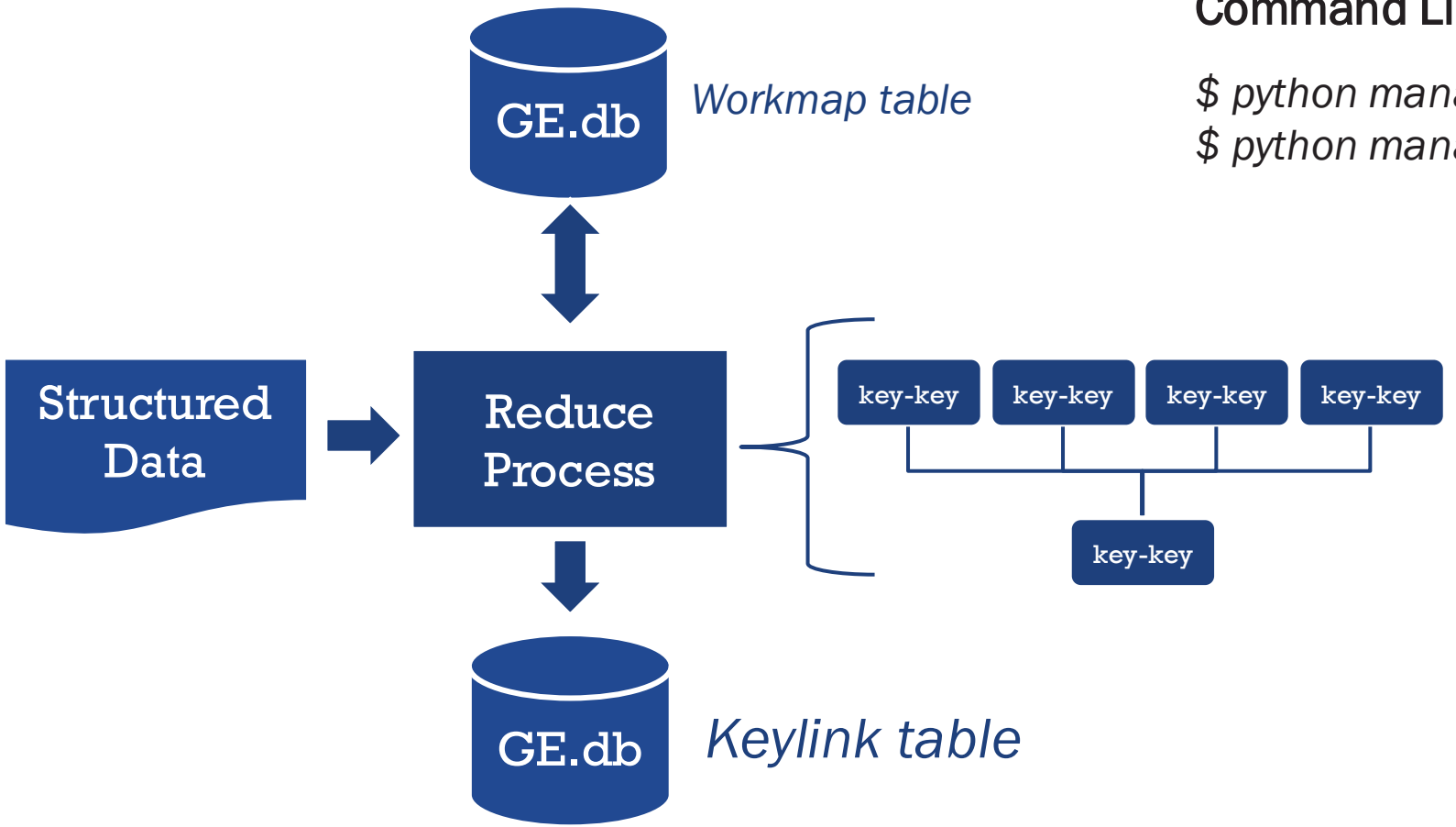
Phase 4 – MapReduce Process



Command Line:

$ python manage.py mapreduce --run {all or dataset}
$ python manage.py mapreduce --reset {all or dataset}

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Workflow Monitor

Allows managing phases and reprocessing

IGEM admin

WELCOME, ANDRE

Home › Ge › Dataset - Workflow

Select wf control to change

Action: -------- Go 0 of 2 selected

DS STATUS	DATASET	LAST UPDATE DATASET	SOURCE FILE VERSION	COLLECT PROCESSED	PREPARE PROCESSED	COMMUTE PROCESSED	MAPREDUCE PROCESSED
False	CTDCGINT	July 6, 2022, 2:10 p.m.	0	✗	✗	✗	✗
True	STRING_CLUSTER	July 6, 2022, 2:20 p.m.	"61f4b7dc-1602a"	✓	✓	✗	✗

2 Dataset - Workflow

allows viewing the 4 phases, with green status indicating successful completion

Control of the last run and the version of the extracted dataset

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Maintanance Processes

Load, download, show, delete, truncate

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db Process

Allows performing maintenance and displaying data from IGEM tables

db –download {table}	*$ python manage.py db –download {table} –path {path_to_write_file}*
db –load {table}	*$ python manage.py db –load {table} –path {path_the_file}*
db –show {table}	*$ python manage.py db –show {table} –{field} {filter_value or all}*
db –delete {table}	*$ python manage.py db –delete {table} –{field} {filter_value or all}*
db –truncate {table}	*$ python manage.py db –truncate {table}*

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DataQuery Processes

Consult and Data Analysis

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Em desenvolvimento

$ python manage.py filter –{table}

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HPC Environment

Run IGEM on ICDS

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Start the System

Script to start Postgres Database

$ module load anaconda3
$ source activate /gpfs/group/mah546/default/sw/igem/igem_env
$ cd /gpfs/group/mah546/default/sw/igem/igem_db
$ pg_ctl -D igemdb -l logfile start

Check Postgres Database status

$ ps aux | grep postgres

Stop Postgres Database

$ kill {id_process)

```
(/gpfs/group/mah546/default/sw/igem/igem_env) [alr6366@comp-bc-0355 src]$ ps aux | grep postgres
alr6366   5775  0.0  0.0 176692 15332 ?        Ss   09:10   0:00 /gpfs/group/mah546/default/sw/igem/igem_
gres -D igemdb
alr6366   5779  0.0  0.0 176808  5088 ?        Ss   09:10   0:00 postgres: checkpointer
alr6366   5780  0.0  0.0 176692  4692 ?        Ss   09:10   0:00 postgres: background writer
alr6366   5781  0.0  0.0 176692  7588 ?        Ss   09:10   0:00 postgres: walwriter
alr6366   5782  0.0  0.0 177240  4372 ?        Ss   09:10   0:00 postgres: autovacuum launcher
alr6366   5783  0.0  0.0  31200  3652 ?        Ss   09:10   0:00 postgres: stats collector
alr6366   5784  0.0  0.0 177244  4152 ?        Ss   09:10   0:00 postgres: logical replication launcher
alr6366  11783  0.0  0.0 178464  7864 ?        Ss   09:19   0:00 postgres: igem gedb ::1(50604) idle
alr6366  22035  0.0  0.0   4788   920 pts/4    S+   09:41   0:00 grep --color=auto postgres
(/gpfs/group/mah546/default/sw/igem/igem_env) [alr6366@comp-bc-0355 src]$ kill 5775
(/gpfs/group/mah546/default/sw/igem/igem_env) [alr6366@comp-bc-0355 src]$
```

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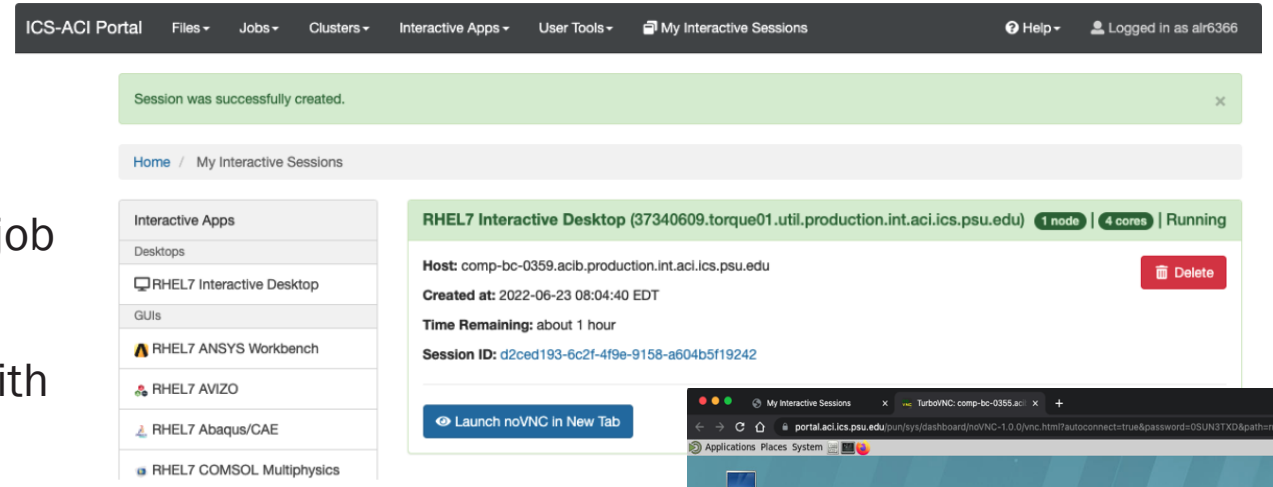
Interface Web

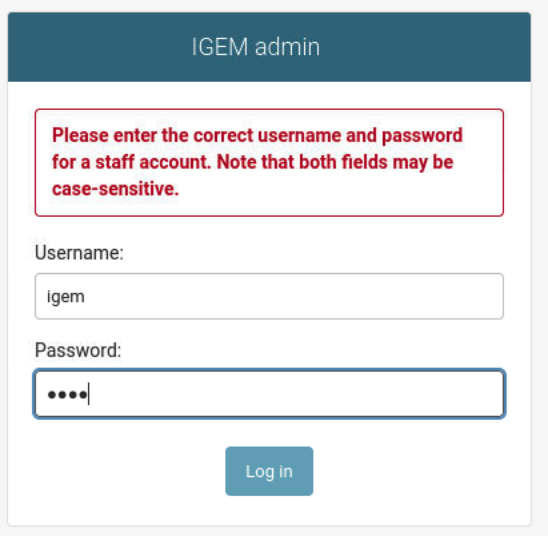
1. Create a RHEL7: interactive Desktop job

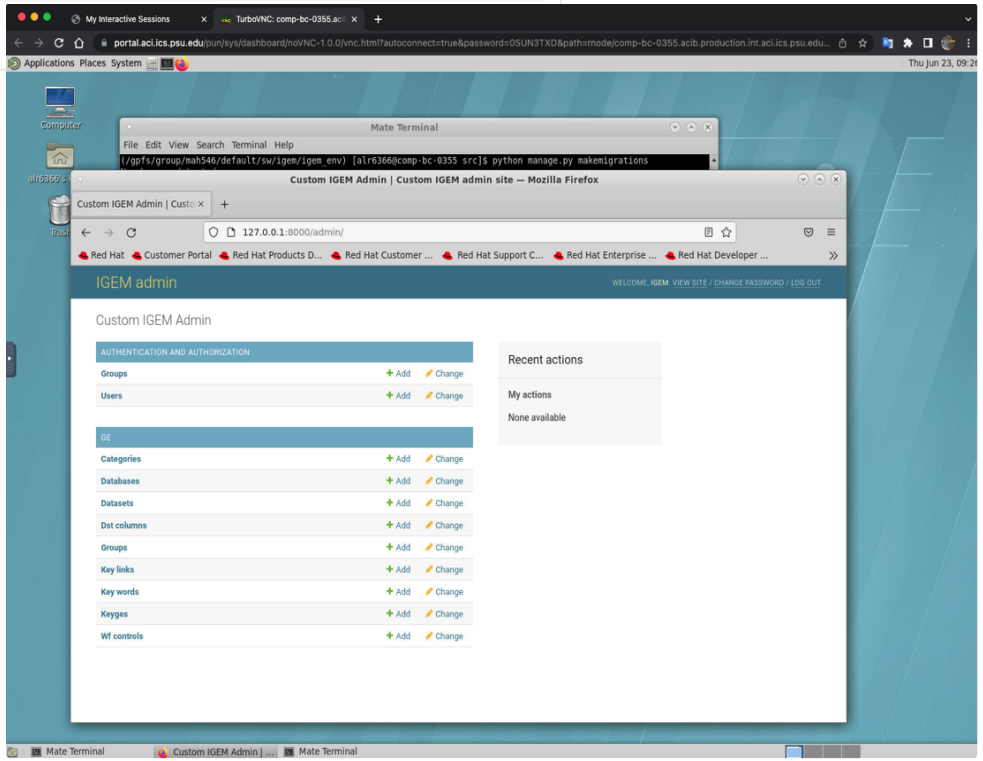
2. On the desktop, run on igem folder with conda environment started: $ python manage.py runserve

3. Open the browser (firefox) and access: http://127.0.0.1:8000/admin

4. Username: igem / password: igem







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JOB to ETL Process

```
#!/bin/bash
#PBS -l nodes=1:ppn=8
#PBS -l walltime=3:00:00
#PBS -l pmem=16gb
#PBS -A open

echo "Job started on `hostname` at `date`"

# Load modules and Activate the environment
module load anaconda3
source activate /gpfs/group/mah546/default/sw/igem/igem_env

# Start Postgres
/gpfs/group/mah546/default/sw/igem/igem_db/pg_ctl -D igemdb -l logfile start

# Run this interactively
/gpfs/group/mah546/default/sw/igem/src/python manage.py collect --run all
/gpfs/group/mah546/default/sw/igem/src/python manage.py prepare --run all
/gpfs/group/mah546/default/sw/igem/src/python manage.py map --run all
/gpfs/group/mah546/default/sw/igem/src/python manage.py reduce --run all

echo "Job Ended at `date`"
```

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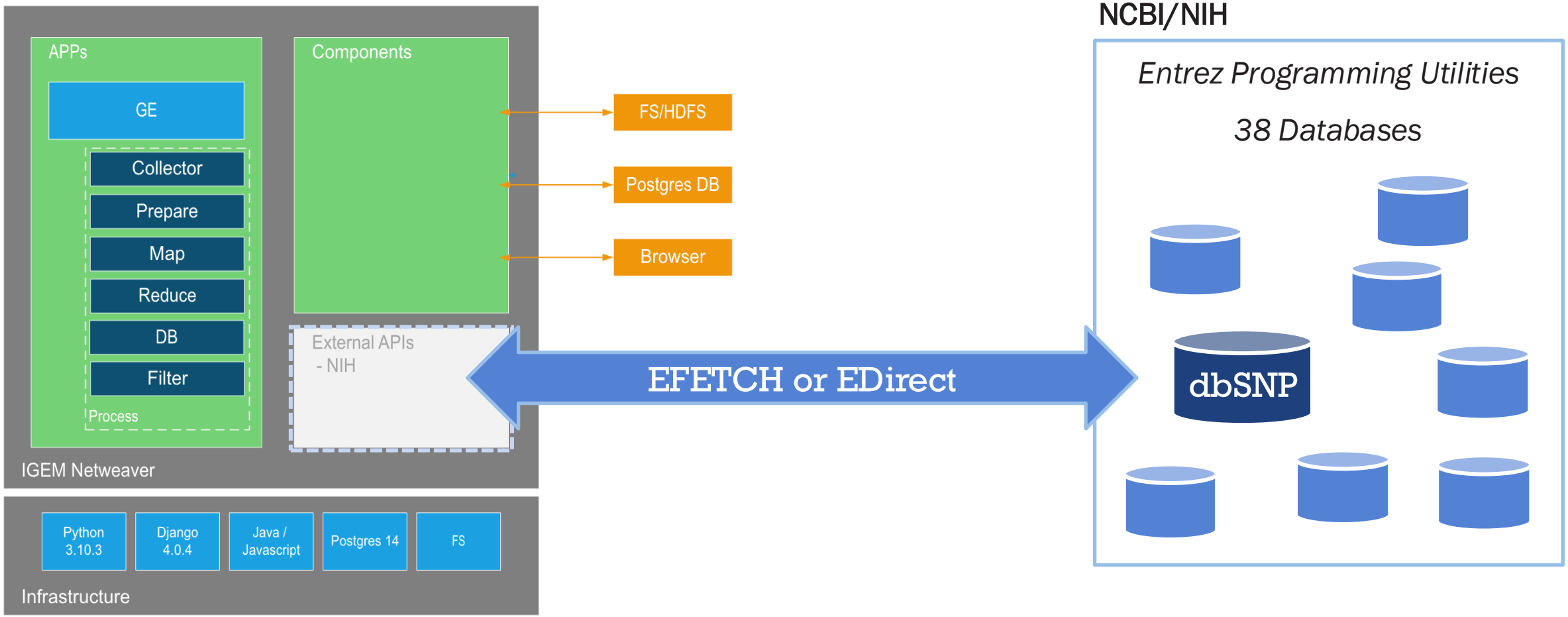
SNPs NIH/Entrez



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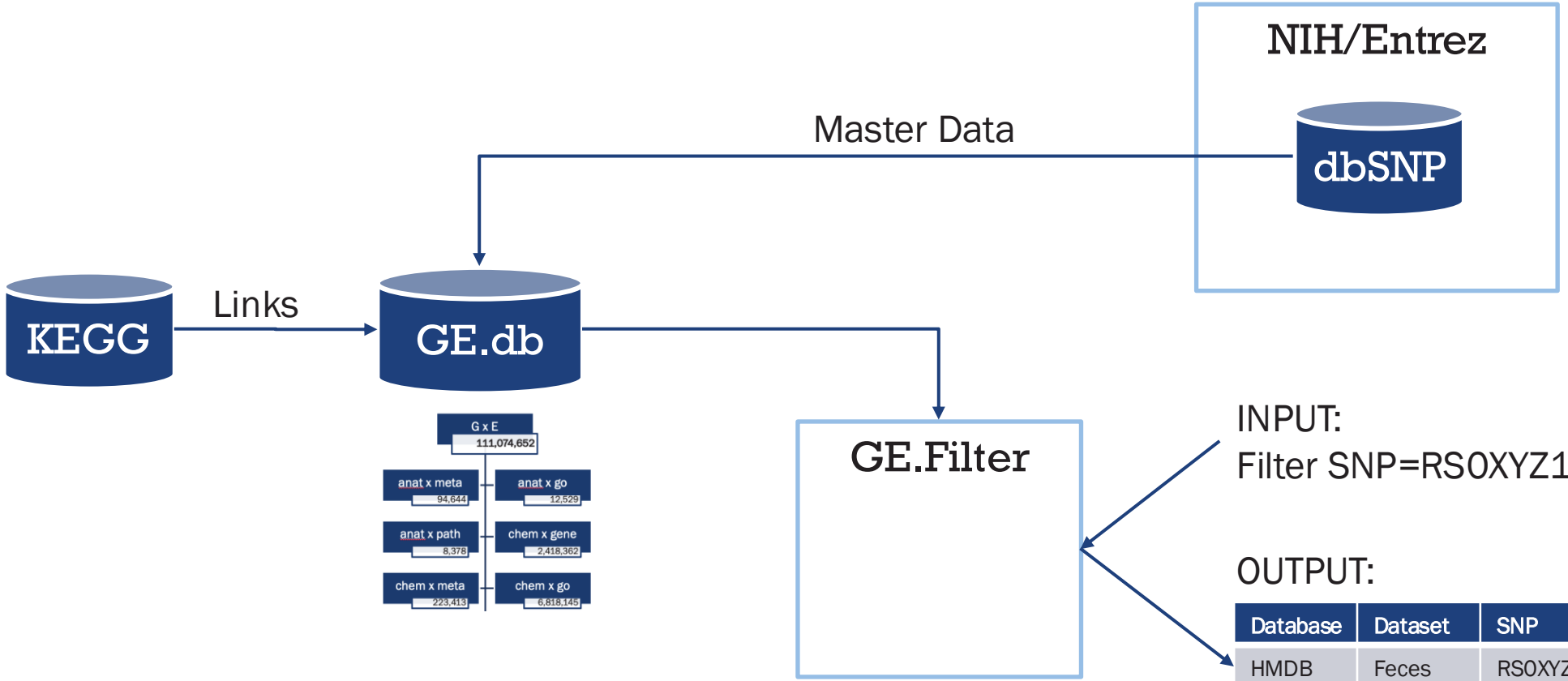
Current Process: Workflow



INPUT:
Filter SNP=RS0XYZ1

OUTPUT:

Database	Dataset	SNP	Exposome	Count
HMDB	Feces	RS0XYZ1	Iron	34
CTD	AssocG	RS0XYZ1	Chemical	3

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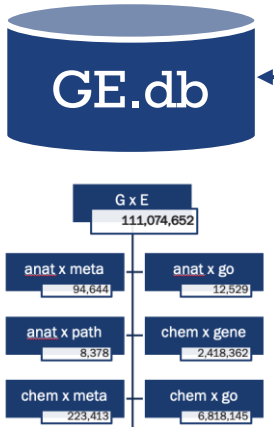
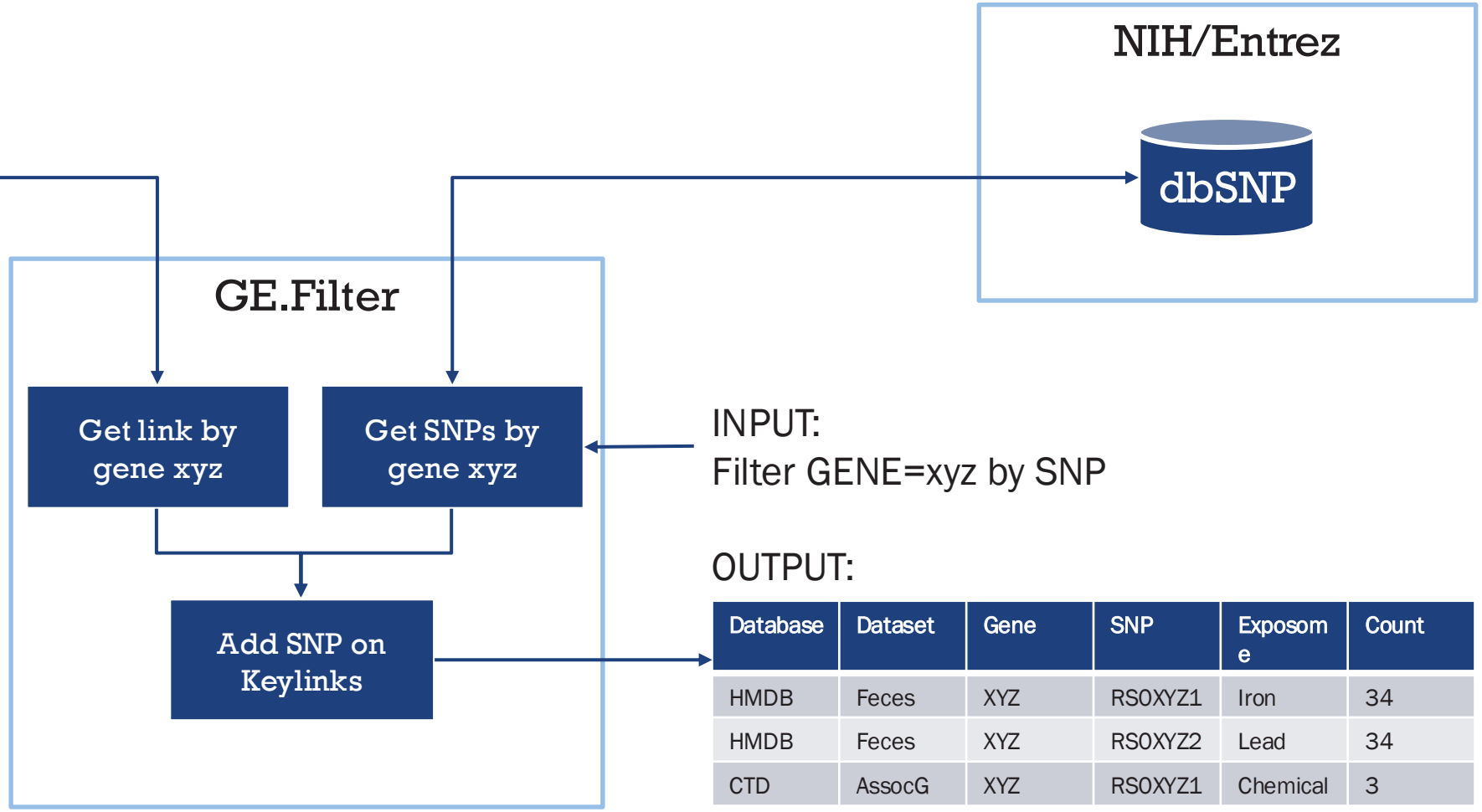
GE.Filter and Entrez: Workflow Example 1





INPUT:
Filter GENE=xyz by SNP

OUTPUT:

Database	Dataset	Gene	SNP	Exposome	Count
HMDB	Feces	XYZ	RS0XYZ1	Iron	34
HMDB	Feces	XYZ	RS0XYZ2	Lead	34
CTD	AssocG	XYZ	RS0XYZ1	Chemical	3

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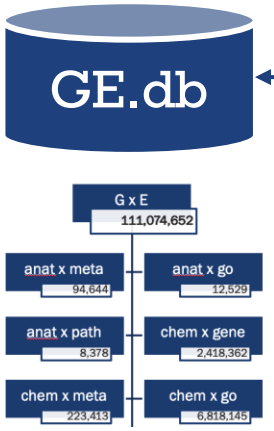
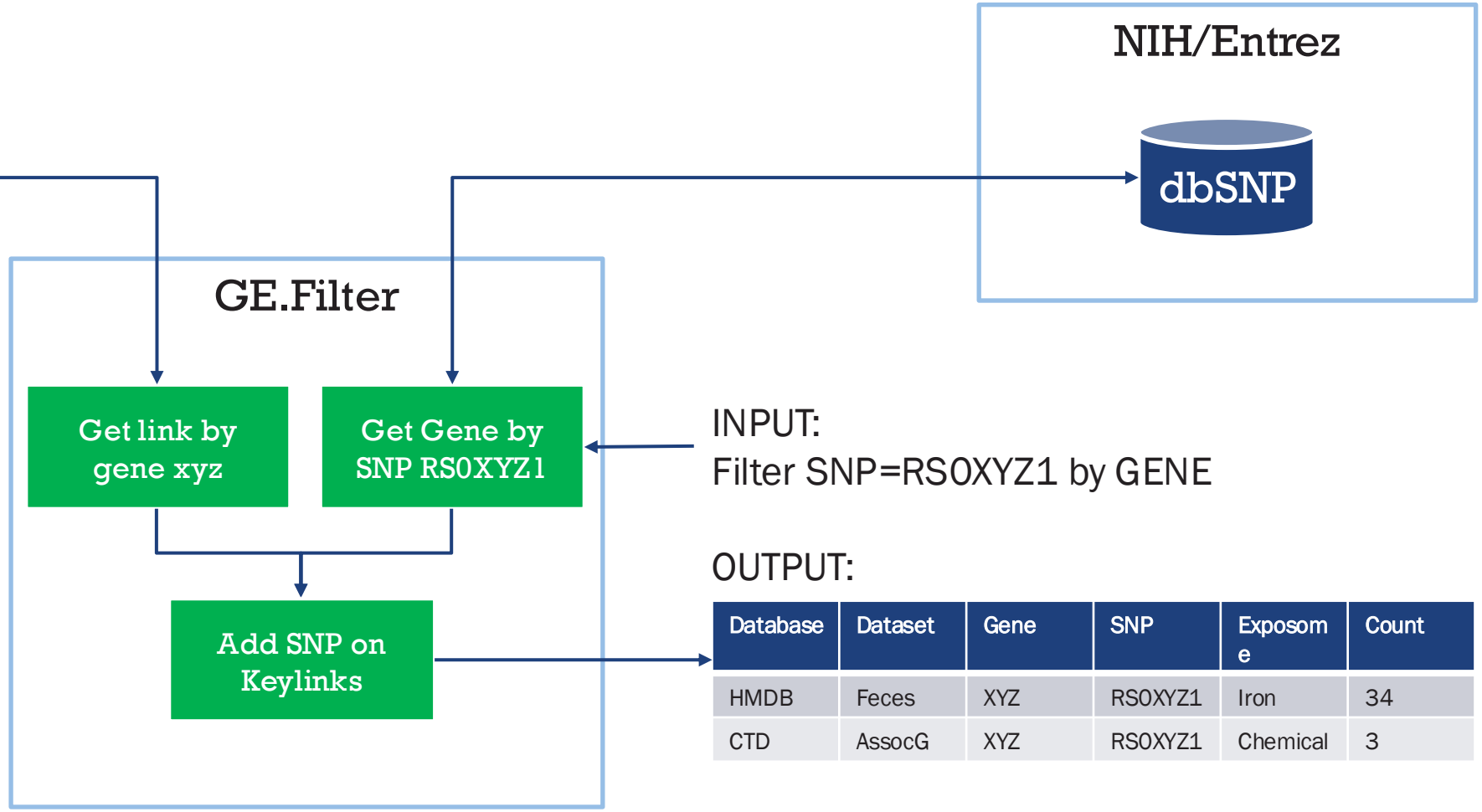
GE.Filter and Entrez: Workflow Example 2





INPUT:
Filter SNP=RS0XYZ1 by GENE

OUTPUT:

Database	Dataset	Gene	SNP	Exposome	Count
HMDB	Feces	XYZ	RS0XYZ1	Iron	34
CTD	AssocG	XYZ	RS0XYZ1	Chemical	3

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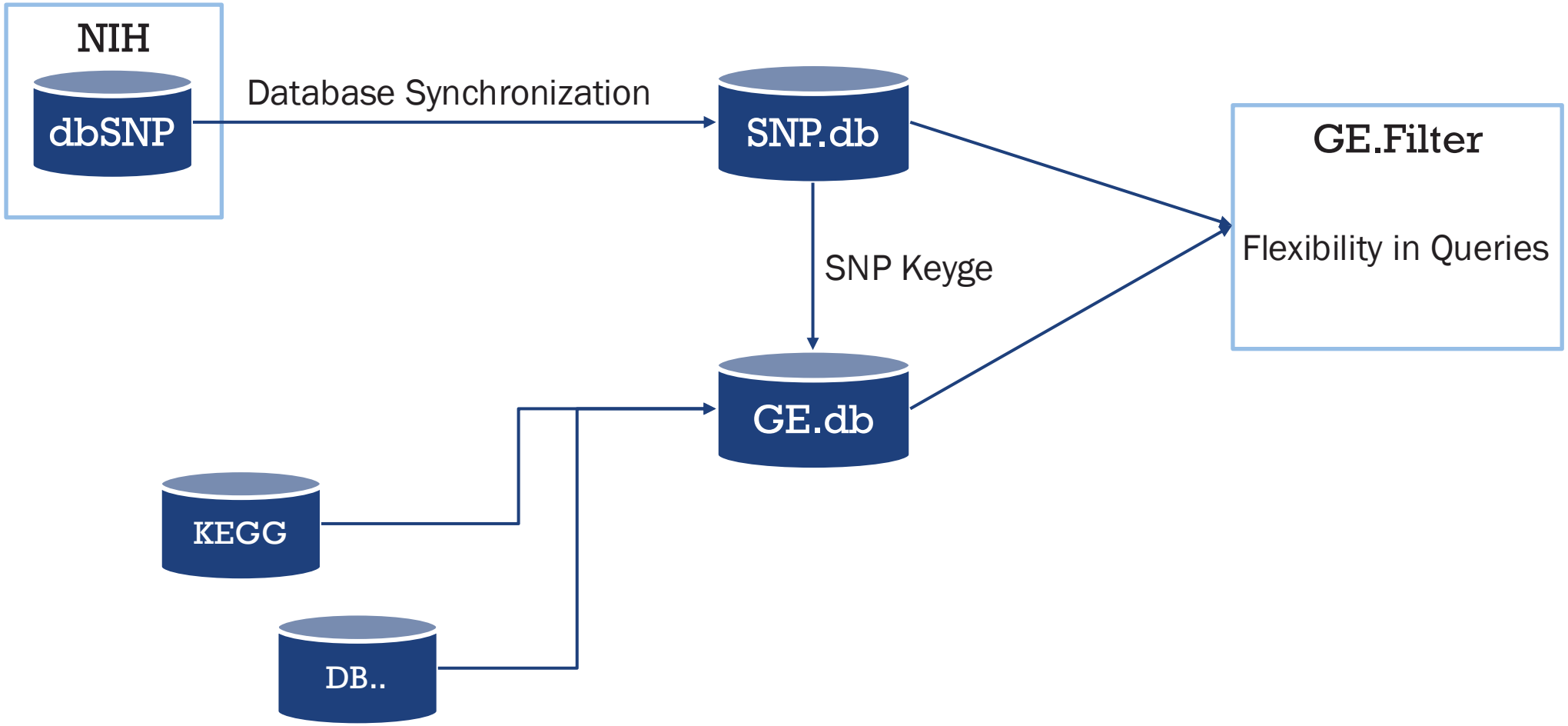
New App



```
● (venv) (base) andrerico@Andres-MacBook-Pro src % python manage.py create_table --run teste

KeyLink:
          geneid        linkid                              linkname   linkcat  count  sources
0    gene:246126  chem:c000228                  aristolochic acid i  chemical      1      1.0
1    gene:246126  chem:c000944                          dicrotophos  chemical      1      1.0
2    gene:246126  chem:c015001                              arsenite  chemical      2      1.0
3    gene:246126  chem:c016805  tris(1,3-dichloro-2-propyl)phosphate  chemical      1      1.0
4    gene:246126  chem:c018475                         butyraldehyde  chemical      1      1.0
..           ...           ...                                   ...       ...    ...      ...
922  gene:246126  dise:d065631                   rhinitis, allergic   disease      1      1.0
923  gene:246126  dise:d065886          neurodevelopmental disorders   disease      2      1.0
924  gene:246126  dise:d066088                         infant death   disease      1      1.0
925  gene:246126  dise:d066126                       cardiotoxicity   disease      2      1.0
926  gene:246126  dise:d066253                  vascular remodeling   disease      2      1.0

[927 rows x 6 columns]

SNPs:
          rsid chrom     start       end       geneid genesymbol
0         3900     y  19568370       nan  gene:246126     txlngy
1         3908     y  19571281       nan  gene:246126     txlngy
2         3909     y  19571275  19571276  gene:246126     txlngy
3         3910     y  19571344       nan  gene:246126     txlngy
4         3911     y  19571567       nan  gene:246126     txlngy
..         ...   ...       ...       ...          ...        ...
311  781205706     y  19603831       nan  gene:246126     txlngy
312  781335843     y  19604113       nan  gene:246126     txlngy
313  878993390     y  19582097  19582098  gene:246126     txlngy
314  879080292     y  19582004       nan  gene:246126     txlngy
315  879146080     y  19586133  19586134  gene:246126     txlngy

[316 rows x 6 columns]

Keylink and SNPs:
             geneid        linkid             linkname   linkcat  count  sources       rsid chrom     start       end genesymbol
0       gene:246126  chem:c000228  aristolochic acid i  chemical      1      1.0       3900     y  19568370       nan     txlngy
1       gene:246126  chem:c000228  aristolochic acid i  chemical      1      1.0       3908     y  19571281       nan     txlngy
2       gene:246126  chem:c000228  aristolochic acid i  chemical      1      1.0       3909     y  19571275  19571276     txlngy
3       gene:246126  chem:c000228  aristolochic acid i  chemical      1      1.0       3910     y  19571344       nan     txlngy
4       gene:246126  chem:c000228  aristolochic acid i  chemical      1      1.0       3911     y  19571567       nan     txlngy
...             ...           ...                  ...       ...    ...      ...        ...   ...       ...       ...        ...
292927  gene:246126  dise:d066253  vascular remodeling   disease      2      1.0  781205706     y  19603831       nan     txlngy
292928  gene:246126  dise:d066253  vascular remodeling   disease      2      1.0  781335843     y  19604113       nan     txlngy
292929  gene:246126  dise:d066253  vascular remodeling   disease      2      1.0  878993390     y  19582097  19582098     txlngy
292930  gene:246126  dise:d066253  vascular remodeling   disease      2      1.0  879080292     y  19582004       nan     txlngy
292931  gene:246126  dise:d066253  vascular remodeling   disease      2      1.0  879146080     y  19586133  19586134     txlngy

[292932 rows x 11 columns]
```

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GE.Filter: non-standard search

Non-standard words

- Hepatitis B core antibody
- Hepatitis B surface antigen
- Hepatitis C antibody (confirmed)
- Hepatitis D (anti-HDV)
- Hepatitis B Surface Antibody
- Hepatitis A Antibody (Anti-HAV)
- HIV antibody test result
- CD4 counts (cells/mm3)
- CD8 counts (cells/mm3)
- Perchlorate, urine (ng/mL)
- Urine perchlorate (ug per g creatinine)
- Water Bromoform (ng/mL)
- Water Chloroform (ng/mL)
- Water Bromodichloromethane (ng/mL)
- Water Dibromochloromethane (ng/mL)
- Water MTBE (ng/mL)
- Blood 1,1-Dichloroethane (ng/mL)
- Blood 1,2-Dichlorobenzene (ng/mL)
- Blood 1,1-Dichloroethene (ng/mL)
- Blood 1,2-Dichloroethane (ng/mL)
- Blood cis-1,2-Dichloroethene (ng/mL)
- Blood trans-1,2-Dichloroethene (ng/mL)
- Blood 1,3-Dichlorobenzene (ng/mL)
- Blood 1,1,2-Trichloroethane (ng/mL)

GE.db

KEYWORD table

KEYLINK table

KEYGE table

Search non-standard words to KEYGE

Search KEYGE

Aggregation Results

OUTPUT

	word [PK] character varying (400)	keyge_id bigint
1	hepatitis b vaccine as04	2038630
2	hepatitis b virus surface antigen p31c	1959928
3	diphtheria-tetanus-acellular pertussis-hib-hepatitis b vaccine	1994854
4	presurface protein 1, hepatitis b surface antigen	1952286
5	presurface protein 2, hepatitis b surface antigen	1952287
6	hepatitis b vaccines	2070991
7	p22 core protein, hepatitis b virus	2040935
8	diphtheria-tetanus-acellular pertussis-inactivated poliovirus-haemophilus influenzae b conjugate-hepatitis b vaccine	2043144
9	hepatitis b antigens	2068522
10	hepatitis b core antigens	2068523
11	hepatitis b e antigens	2068524
12	hepatitis b surface antigens	2068525
13	hepatitis b	2077984
14	hepatitis b, chronic	2077985

	ckey [PK] character varying (15)	count integer	dataset_id bigint	keyge1_id bigint	keyge2_id bigint
1	397-571	1	397	1897281	2077984
2	397-4219	1	397	1914628	2077984
3	397-20554	4	397	2066530	2077984
4	397-25330	1	397	2067911	2077984
5	397-27375	4	397	2068052	2077984
6	397-28409	2	397	1941799	2077984
7	397-35729	1	397	2070329	2077984
8	397-35877	20	397	2069760	2077984
9	397-36106	1	397	1940072	2077984

GE.Filter: non-standard search

```
11    "Perchlorate, urine (ng/mL)"
```

Search:
Urine – 658 rows
Perchlorante – 66 rows
Urine, perchlorante – 0 rows

	word [PK] character varying (400)	keyge_id bigint	commute boolean	status boolean
1	urine	1897120	true	true
2	d014556	1897120	true	true
3	anat:d014556	1897120	false	true
4	perchlorate	2018389	true	true
5	c494474	2018389	true	true
6	chem:c494474	2018389	false	true
7	creatinine	2655556	true	true
8	hmdb0000562	2655556	true	true
9	meta:hmdb0000562	2655556	false	true

```
11    9,chem:c494474 meta:hmdb0000562 anat:d014556
```

	id [PK] bigint	keyge character varying (40)	description character varying (400)	category_id bigint	group_id bigint
1	1897120	anat:d014556	urine	14	2
2	2018389	chem:c494474	perchlorate	13	2
3	2655556	meta:hmdb0000562	creatinine	16	1

IGEM Word Search

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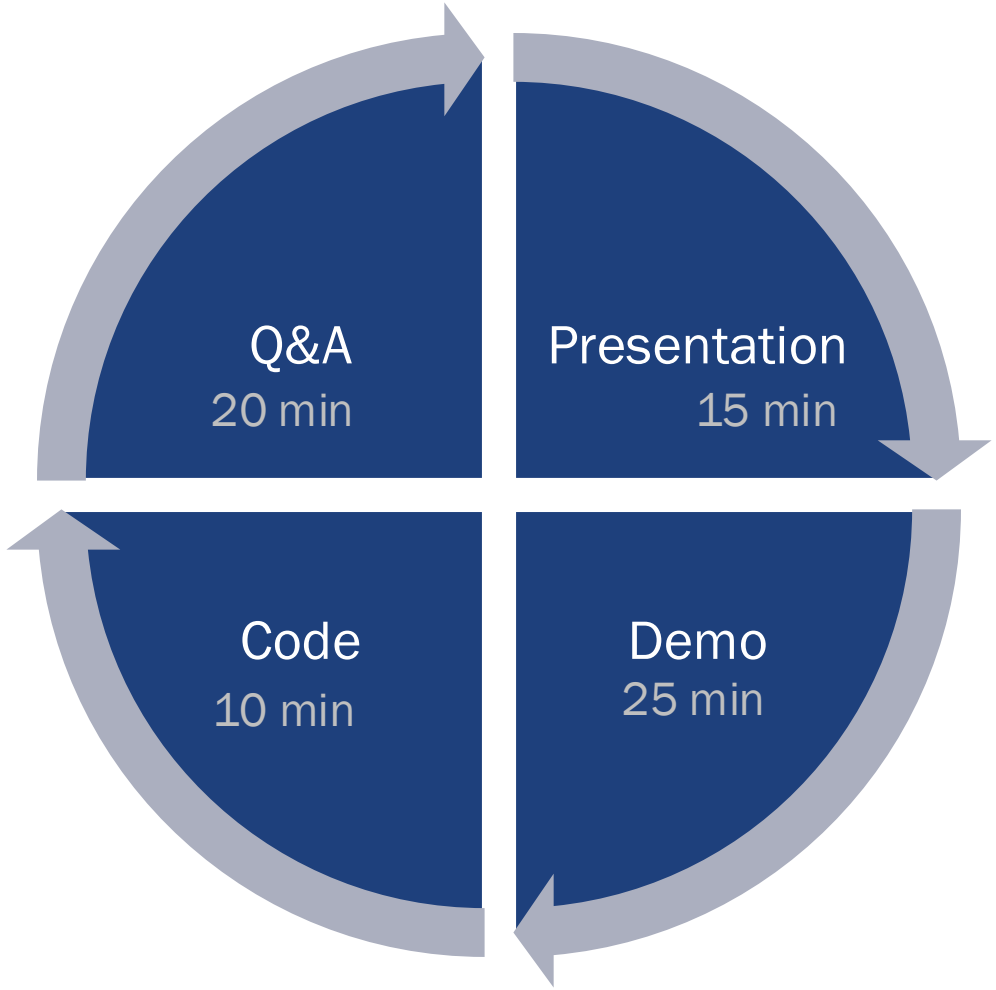
Content



theoretically

Points we will address:

- How IGEM performs the search for words and converts to terms (keyge)
- How to start an IGEM installation
- How to enter data for search
- How to perform the search

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Database

DATABASE table

DATASET table

DSTCOLUMN table

AUTH_USER table

ADMIN_LOG table

SNPGENE table

AUTH_PERMISSION table

WFCONTROL table

AUTH_GROUP table

GE.db

PREFIXOPC table

WORDMAP table

ADMIN_SESSION table

LOGCOLLECTOR table

KEYHIERARCHY table

AUTH_USER_GROUP table

KEYLINK table

GROUP table

CATEGORY table

KEYWORD table

- Stores all links between external words with system terms (keyge).
- The table will be the intelligence of the system, improving with use and interactions.
- Data entry will be manual and may be automated in the future by an AI algorithm.

Fields: word / keyge / activate / commute

KEYGE table

- Stores the master data of the terms (keygen).
- It will be sorted by groups and categories.

fields: ID / keyge / description / group / category

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Search Flow

	A
1	var_desc
2	Hepatitis B core antibody
3	Hepatitis B surface antigen
4	Hepatitis C antibody (confirmed)
5	Hepatitis D (anti-HDV)
6	Hepatitis B Surface Antibody
7	Hepatitis A Antibody (Anti-HAV)
8	HIV antibody test result
9	CD4 counts (cells/mm3)
10	CD8 counts (cells/mm3)
11	Perchlorate, urine (ng/mL)
12	Urine perchlorate (ug per g creatinine)

Separate each word

hepatitis
b
core
antibody

search each word individually

```
1  select * from ge_keyword where word like '%hepatitis%'
```

Data output Messages Notifications

	word [PK] character varying (400)	keyge_id bigint
1	diphtheria-tetanus-acellular pertussis-hib-hepatitis b vaccine	1994854
2	presurface protein 1, hepatitis b surface antigen	1952286
3	presurface protein 2, hepatitis b surface antigen	1952287
4	hepatitis b virus surface antigen p31c	1959928
5	c100 protein, hepatitis c virus	1962439
6	core protein p9, hepatitis c virus	2051785
7	core protein p22, hepatitis c virus	2019995
8	hepatitis vaccine mth 68-b	1995882
9	hepatitis delta virus large antigen	2004616
10	hepatitis b vaccine as04	2038630
11	p22 core protein, hepatitis b virus	2040935
12	diphtheria-tetanus-acellular pertussis-inactivated poliovirus-haemophilus influenzae b conjugate-hepatitis b vaccine	2043144
13	core protein p13, hepatitis c virus	2051786
14	hepatitis b antigens	2068522
15	halothane hepatitis	2055027
16	hepatitis b core antigens	2068523
17	hepatitis b e antigens	2068524
18	hepatitis b surface antigens	2068525
19	hepatitis	2077982
20	viral hepatitis vaccines	2070589
21	hepatitis b vaccines	2070991
22	hepatitis c antigens	2071202
23	hepatitis antigens	2071215
24	hepatitis a vaccines	2071501
25	hepatitis a antigens	2071588
26	hepatitis delta antigens	2071589

Total rows: 41 of 41 Query complete 00:00:00.652

Unifies all results for each search word into a single list

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Search Flow

Sort the list from largest words to smallest

Unifies all results for each search word into a single list

word
diphtheria-tetanus-acellular pertussis-inactivated poliovirus-haemophilus influenzae b conj
diphtheria-tetanus-acellular pertussis-hib-hepatitis b vaccine
presurface protein 1, hepatitis b surface antigen
presurface protein 2, hepatitis b surface antigen
hepatitis b virus surface antigen p31c
core protein p22, hepatitis c virus
hepatitis delta virus large antigen
p22 core protein, hepatitis b virus
core protein p13, hepatitis c virus
core protein p9, hepatitis c virus
c100 protein, hepatitis c virus
hepatitis b surface antigens
hepatitis, infectious canine
hepatitis vaccine mth 68-b
hepatitis b core antigens
hepatitis b vaccine as04
viral hepatitis vaccines
hepatitis delta antigens
hepatitis, viral, animal
hepatitis, viral, human
hepatitis b e antigens
hepatitis, autoimmune
hepatitis b antigens
hepatitis b vaccines
hepatitis c antigens
hepatitis a vaccines
hepatitis a antigens
hepatitis b, chronic
hepatitis c, chronic
hepatitis d, chronic
hepatitis, alcoholic
halothane hepatitis
hepatitis antigens
hepatitis, chronic
hepatitis, animal
hepatitis a
hepatitis b
hepatitis c
hepatitis d
hepatitis e
hepatitis

first match

Hepatitis B core antibody

second match

Core antibody

attempts to finish the list

found terms

keyge xyz, keyge acm

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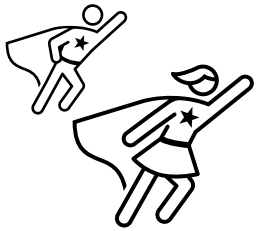
DEMO



Let`s see this in real life

1. Git Clone
2. Configuring the Python environment
3. Configuring the database
4. configuring user
5. Configuring master data
6. Run the first search

IGEM Simple Example

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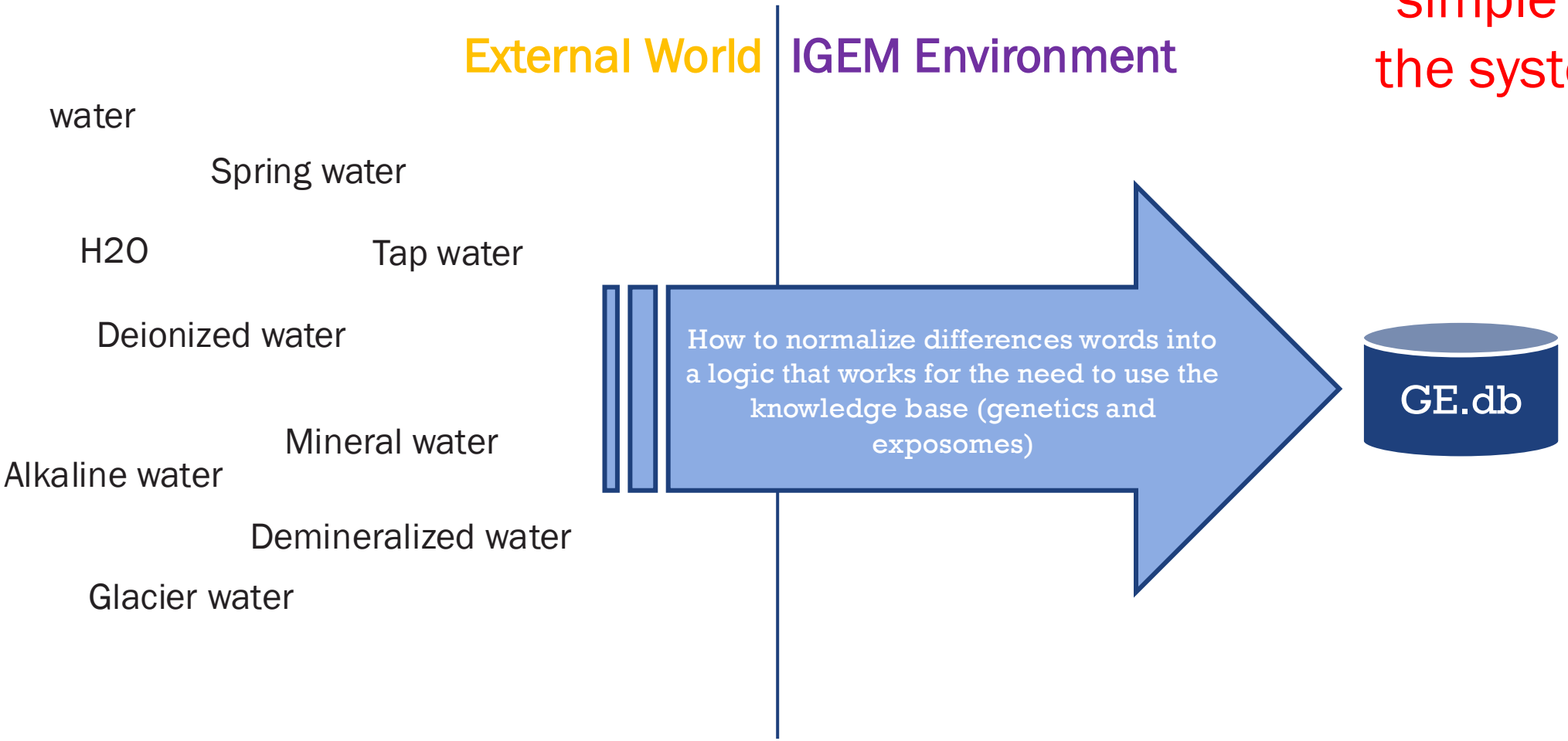
Many databases and many words

Example scenario to illustrate in a simple way how the system works



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Step 1: Identify what to look for

Example scenario to illustrate in a simple way how the system works

External World

water

Spring water

H2O

Tap water

Deionized water

Mineral water

Alkaline water

Demineralized water

Glacier water

IGEM Environment

Keyword (Dictionary)

External Words	Keyge (terms)
water	water
H2O	water
tap water	water
spring water	spring water
mineral water	spring water
Glacier water	spring water
Deionized water	Industrial water
Desmineralized water	Industrial water
Alkaline water	Special water

Today this process is manual, but with the use and improvement of the algorithm, we will have an A.I. to automate this normalization

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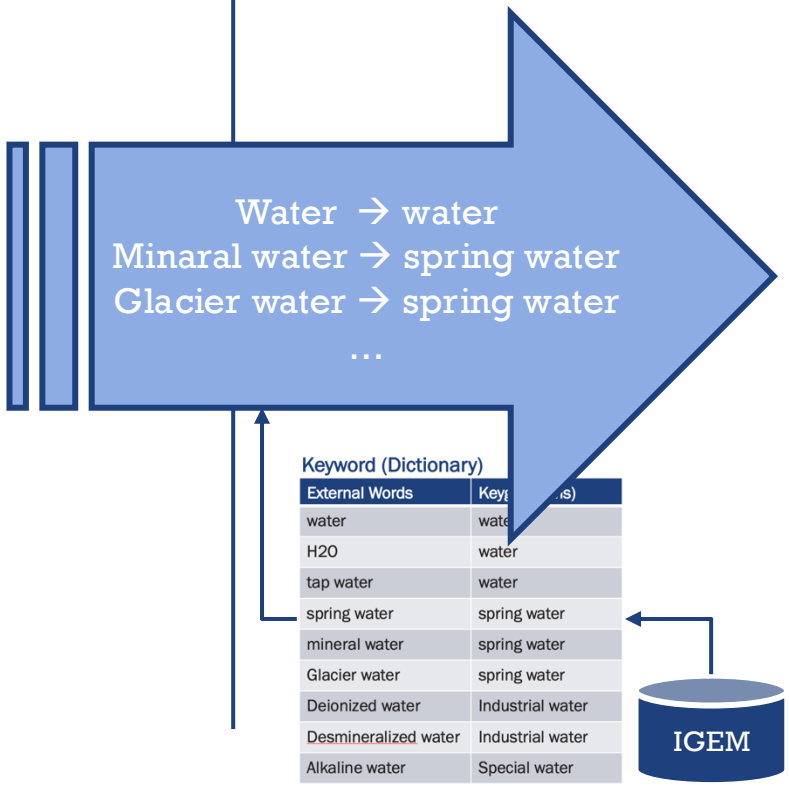
Step 2: Normalize out words to IGEM standard

Example scenario to illustrate in a simple way how the system works

External World | IGEM Environment

Field A	Field B	Field C
water	apple	lake
H2O	water	Ocean
tap water	Spring water	Alkaline water
tap water	Mineral water	5.2
mineral water	spring water	9
Glacier water	pineapple	Water from
Deionized water	Green Water	8
Desmineralized water	Water of mart	5.2
Alkaline water	Special water	Water tank



Field A	Field B	Field C
water	apple	lake
water	water	Ocean
water	Spring water	Special water
water	spring water	5.2
spring water	spring water	9
spring water	pineapple	water from
industrial water	Green Water	8
industrial water	water of mart	5.2
special water	Special water	water tank

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Step 3: MAPPER

Example scenario to illustrate in a simple way how the system works

IGEM Environment

	Field A	Field B	Field C
one term found, no mapping	water	apple	lake
same term found, no mapping	water	water	Ocean
	water	Spring water	Special water
	water	spring water	5.2
same term found, no mapping	spring water	spring water	9
	spring water	pineapple	water from
	industrial water	Green Water	8
	industrial water	water of mart	5.2
	special water	Special water	water tank

Keyge 1	Keyge 2
water	Spring water
water	Special water
Special water	Spring water
Water	Spring water
Spring water	water
Industrial water	water
Industrial water	water
Special water	water

IGEM ETL to CTD Datasets

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XXX

Example scenario to illustrate in a simple way how the system works

Field A	Field B	Field C
water	apple	lake
water	water	Ocean
water	Spring water	Special water
water	spring water	5.2
spring water	spring water	9
spring water	pineapple	water from
industrial water	Green Water	8
industrial water	water of mart	5.2
special water	Special water	water tank

ETL Rule
Field A – Field C

Keyge 1	Keyge 2	Count	Souces
water	Special water	2	1
water	Spring water	1	1

IGEM GE Database

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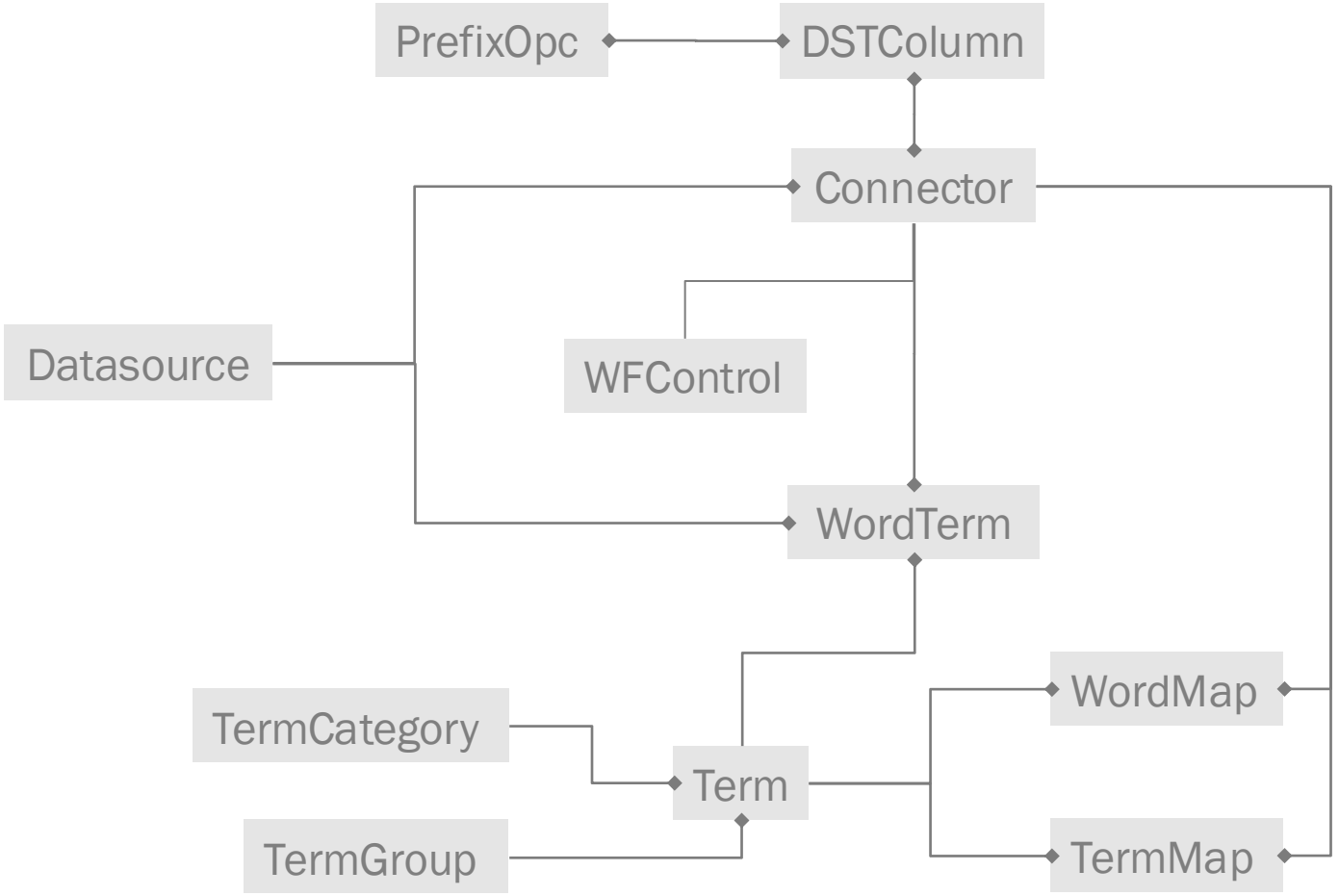
Database Diagram



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Database GE: Datasource table

Datasource table

- id
- datasource
- description
- category
- website

Goal: It stores the external data sources with their respective addresses on the WEB and their classification. The Connectors will use the Data Sources to group an information domain.

- Connector table
- WordMap table

Functions in python:

```
from igem.ge import db
db.get_data()
db.load_data()
db.delete()
db.truncate()

db.get_data(table="datasource",
    datasource={"datasource__in": ["ds_01","ds_02"]},
    columns=["id","datasource"],
    columns_out=["Datasource ID", "Datasource Name"],
    path="{your_path}/datasource.csv")
```

Table Example

	id [PK] bigint	datasource character varying (20)	description character varying (200)	category character varying (20)	website character varying (200)
1	12	hmdb	human metabolome database	genomics	https://hmdb.ca/
2	13	ctd	comparative toxicogenomics database	genomics	http://ctdbase.org/
3	14	digchem	digchem	genomics	http://gcancer.org/digchem
4	15	digsee	digsee	genomics	http://gcancer.org/digsee

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Database GE: Connector table

GE.db

Datasource table

Connector table

- id
- connector
- datasource_id (fk)
- description
- update_ds
- source_path
- source_web
- source_compact
- source_file_name
- source_file_format
- source_file_sep
- source_file_skiprow
- target_file_name
- target_file_format
- target_file_keep

Goal: Stores extract data from sources external to GE.db as well as connection parameters.

- DSTColumn table
- WordMap table
- TermMap table
- WFControl table
- LogsCollector table

Functions in python:

from igem.ge import db
db.get_data()
db.load_data()
db.delete()
db.truncate()

db.get_data(table="connector",
datasource={"datasource_id__in": ["ds_01","ds_02"]},
connector={"connector__in":["ds_01_conn_01"]
columns=["id","datasource"],
columns_out=["Datasource ID", "Datasource Name"],
path="{your_path}/datasource.csv")

If the path parameter is informed, a file will be generated, and the return of the get_data function will be a boolean. Otherwise, the function returns a DataFrame with the selected data.

Table Example

	id [PK] bigint	connector character varyin	description character varying (200)	update_ds boolean	source_path character varying (300)	source_web boolean	source_comp boolean	source_file_name character varying (200)	source_file_format character varying (200)	source_file_sep character varying (3)	source_file_skiprow integer	target_file_name character varying (200)	target_file_format character varying (200)	target_file_keep boolean	datasource_id bigint
9	9	ctdcphasso	Chemicalphenotype interactions	false	http://ctdbase.org/reports/CTD_pheno_term_ixns.csv.gz	true	true	CTD_pheno_term_ixns.csv.gz	GZ	,	0	CTD_pheno_term_ixns.csv	CSV	false	13
10	10	ctdesassoc	Exposurestudy associations	false	http://ctdbase.org/reports/CTD_exposure_studies.csv.gz	true	true	CTD_exposure_studies.csv.gz	GZ	,	0	CTD_exposure_studies.csv	CSV	false	13
11	11	ctdeeassoc	Exposureevent associations	false	http://ctdbase.org/reports/CTD_exposure_events.csv.gz	true	true	CTD_exposure_events.csv.gz	GZ	,	0	CTD_exposure_events.csv	CSV	false	13
12	12	ctdpgodaso	Phenotype (GO)Disease Biological Process...	false	http://ctdbase.org/reports/CTD_Phenotype-Disease_biological_proces...	true	true	CTD_Phenotype-Disease_biological_proc...	GZ	,	0	CTD_Phenotype-Disease_biological_process_associations.c...	CSV	false	13
13	13	ctdpgodcca	Phenotype (GO)Disease Cellular Compone...	false	http://ctdbase.org/reports/CTD_Phenotype-Disease_cellular_compone...	true	true	CTD_Phenotype-Disease_cellular_compo...	GZ	,	0	CTD_Phenotype-Disease_cellular_component_associations....	CSV	false	13
14	14	ctdpgodmfa	Phenotype (GO)Disease Molecular Functio...	false	http://ctdbase.org/reports/CTD_Phenotype-Disease_molecular_functio...	true	true	CTD_Phenotype-Disease_molecular_fun...	GZ	,	0	CTD_Phenotype-Disease_molecular_function_associations....	CSV	false	13

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Database GE: PrefixOpc table

PrefixOpc table

- pre_value

Goal: Some of the data ingested by GE.db are codes of genes, chemicals, and others. These codes are not unique and may have overlapping. To keep the GE terms in line with the external regulations and to help with identification, we add a prefix to the code and store it as a term. This association will be performed when configuring the Connector's columns.

DSTColumn table

Functions in python:

from igem.ge import db
db.get_data()
db.load_data()
db.delete()
db.truncate()

Df = db.get_data(table="prefix")

Table Example

	pre_value [PK] character varying (5)
1	chem:
2	dise:
3	gene:
4	go:
5	none

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Database GE: DSTColumn table

PreFixOpc table

Connector table

DSTColumn table

- Id **PK**
- connector_id **FK**
- status
- column_number
- column_name
- pre_value **FK**
- single_word

Goal: Stores the parameters for ingesting the data provided by the connector. Parameters like: if a column should be processed or not if the column is a code or a text.

Functions in python:

```
from igem.ge import db
db.get_data()
db.load_data()
db.delete()
db.truncate()

Df = db.get_data(
   table="ds_column",
   connector={"connector_id__connector__in": "ds_01_conn_01"}
   )
```

Table Example

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Database GE: WFControl table

connector table

WFControl table

- Id **PK**
- connector_id **FK**
- Last_update
- Source_file_version
- Source_file_size
- Target_file_size
- Chk_collect
- Chk_prepare
- Chk_map
- Chk_reduce

Goal:

Functions in python:

```
from igem.ge import db
db.get_data()
db.load_data()
db.delete()
db.truncate()

Df = db.get_data(
   table="workflow",
   connector={"connector_id__connector__in": "ds_01_conn_01"}
   )
```

Table Example

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Database GE: TermGroup table

GE.db

TermGroup table

- id PK
- term_group
- description

Goal:

Term table

Functions in python:

```
from igem.ge import db
db.get_data()
db.load_data()
db.delete()
db.truncate()

Df = db.get_data(
   table="term_group",
   )
```

Table Example

	id [PK] bigint	term_group character varying (20)	description character varying (200)
1	1	genomic	Genomic Group
2	2	environment	Environment Word Group

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Database GE: TermCategory table

GE.db

TermCategory table

- id PK
- term_category
- description

Goal:

Term table

Functions in python:

from igem.ge import db
db.get_data()
db.load_data()
db.delete()
db.truncate()

Df = db.get_data(
table="term_group",
)

Table Example

	id [PK] bigint	term_category character varying (20)	description character varying (200)
1	4	metal	Metal words category
2	5	gene	Gene words category
3	6	disease	Disease words category
4	7	snp	SNPs words category
5	8	chromosomes	Chromosomes words category
6	10	taxonomy	Taxonomy Category
7	11	cluster	cluster
8	13	chemical	chemical
9	14	anatomy	anatomy
10	15	pathway	pathway
11	16	metabolite	metabolites
12	18	ontology	Gene Ontology

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Database GE: Term table

TermGroup table

TermCategory table

Term table

- id PK
- term
- description
- term_group_id FK
- term_category_id FK

Goal:

TermMap table

WordMap table

WordTerm table

Functions in python:

```
from igem.ge import db
db.get_data()
db.load_data()
db.delete()
db.truncate()

Df = db.get_data(
    table="term",
    term_group={"term_group_id__term_group__in":["genomic"]}
    )
```

Table Example

	id [PK] bigint	term character varying (40)	description character varying (400)	term_category_id bigint	term_group_id bigint
1	1895361	anat:d016475	3t3 cells	14	2
2	1895362	anat:d041721	3t3-l1 cells	14	2
3	1895363	anat:d000072283	a549 cells	14	2
4	1895364	anat:d000005	abdomen	14	2
5	1895365	anat:d034841	abdominal cavity	14	2
6	1895366	anat:d000090043	abdominal core	14	2

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Database GE: WordTerm table

WordTerm table

- word PK
- term_id FK
- status
- commute

Goal:

TermGroup table

TermCategory table

Functions in python:

```
from igem.ge import db
db.get_data()
db.load_data()
db.delete()
db.truncate()

Df = db.get_data(
   table="wordterm",
   term_group={"term_id__term_group_id__term_group__in":["genomic"]}
   )
```

Table Example

	word [PK] character varying (400)	status boolean	commute boolean	term_id bigint
17	(r)-6-((s)-2-hydroxy-4-(4-hydroxyphenyl)butyl)-5,6-dihydropyran-2-one	true	true	2036329
18	(((((4-hydroxy-4,4-diphosphonobutyl)carbamoylmethyl)carbamoylmethyl)carbamoylmethyl)carbamoylmethanethiolate)oxorhenium(v)	true	true	2029935
19	(((1,2-5,6-di-o,o-isopropylideneglucofuranos-3-yl)oxocarbonyl)methyl)cobalt tricarbonyl triphenylphosphine	true	true	2003131
20	(((1,3-dibenzyl-2,5-dioxo-4-imidazolidinylidene) methyl)sulfanyl) methyl acetate	true	true	2039725
21	(((2-(2-(2,4-dichlorophenyl)-4-(2-methyl-3-chlorophenyl)-1-ylmethyl)carbamoyl)methyl)-5-oxopyrrole-1-yl)-n-(3-piperidinylpropyl)acetamide	true	true	2035034

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Database GE: WordMap table

Datasource table

Connector table

Term table

WordMap table

- cword PK
- datasource_id FK
- connector_id FK
- term_1_id FK
- term_2_id FK
- word_1
- word_2
- qtd_links

Goal:

Functions in python:

```
from igem.ge import db
db.get_data()
db.load_data()
db.delete()
db.truncate()

Df = db.get_data(
   table="wordmap",
   term_group={"term_1_id__term_group_id__term_group__in":["genomic"]}
   )
```

Table Example

	cword [PK] character varying (15)	word_1 character varying (100)	word_2 character varying (100)	qtd_links integer	connector_id bigint	datasource_id bigint	term_1_id bigint	term_2_id bigint
1	1	dsds	sdfsdfg	5	1	13	2490758	1895361

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Database GE: TermMap table

GE.db

TermMap table

- ckey PK
- connector_id FK
- term_1_id FK
- term_2_id FK
- qtd_links

Goal:

Connector table

Term table

Functions in python:

```
from igem.ge import db
db.get_data()
db.load_data()
db.delete()
db.truncate()

Df = db.get_data(
   table="termmap",
   term_group={"term_1_id__term_group_id__term_group__in":["genomic"]}
   )
```

Table Example

	ckey [PK] character varying (15)	qtd_links integer	connector_id bigint	term_1_id bigint	term_2_id bigint
1	4-103001	1	4	1922212	2490758
2	4-103002	1	4	1922212	2167723
3	4-103003	1	4	1922212	2167778

IGEM Echo Report

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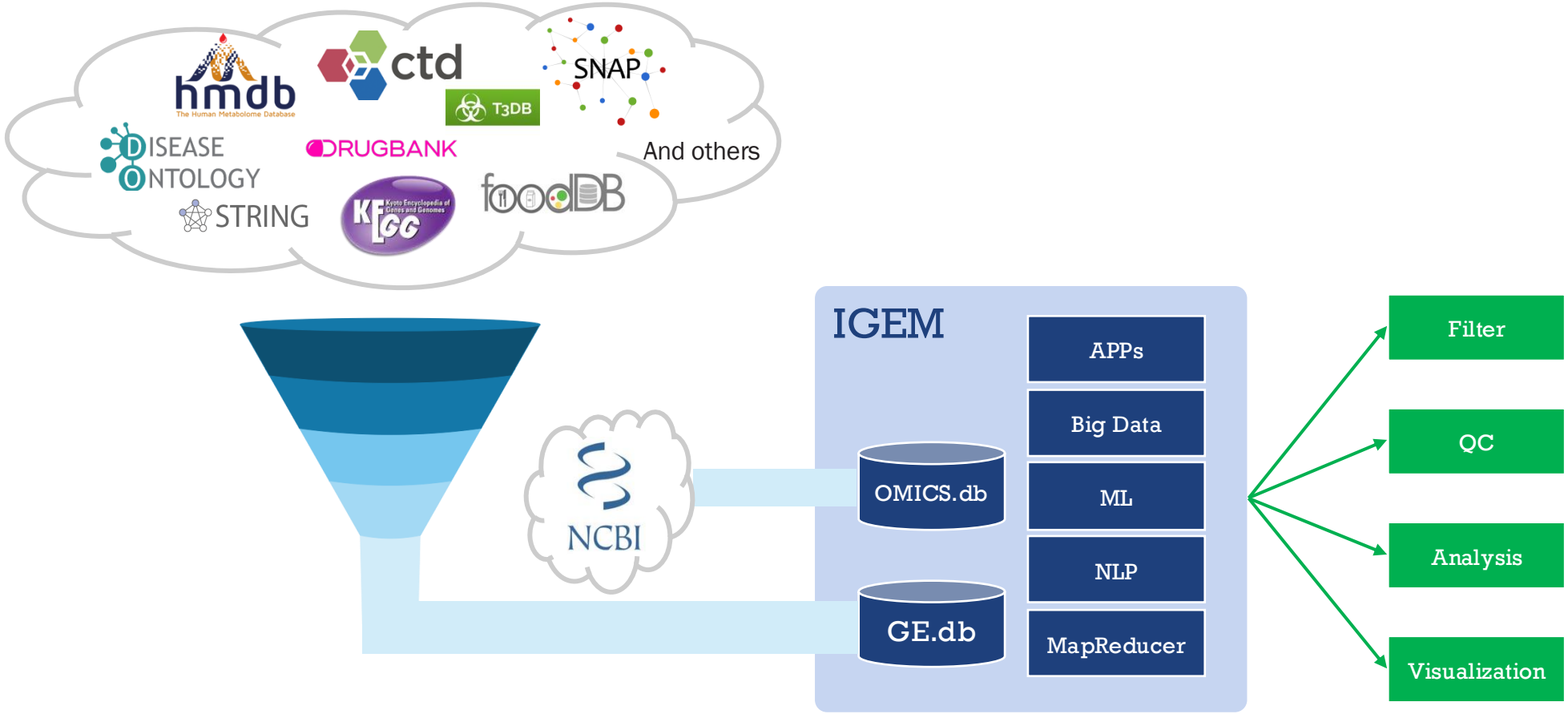
IGEM Overview



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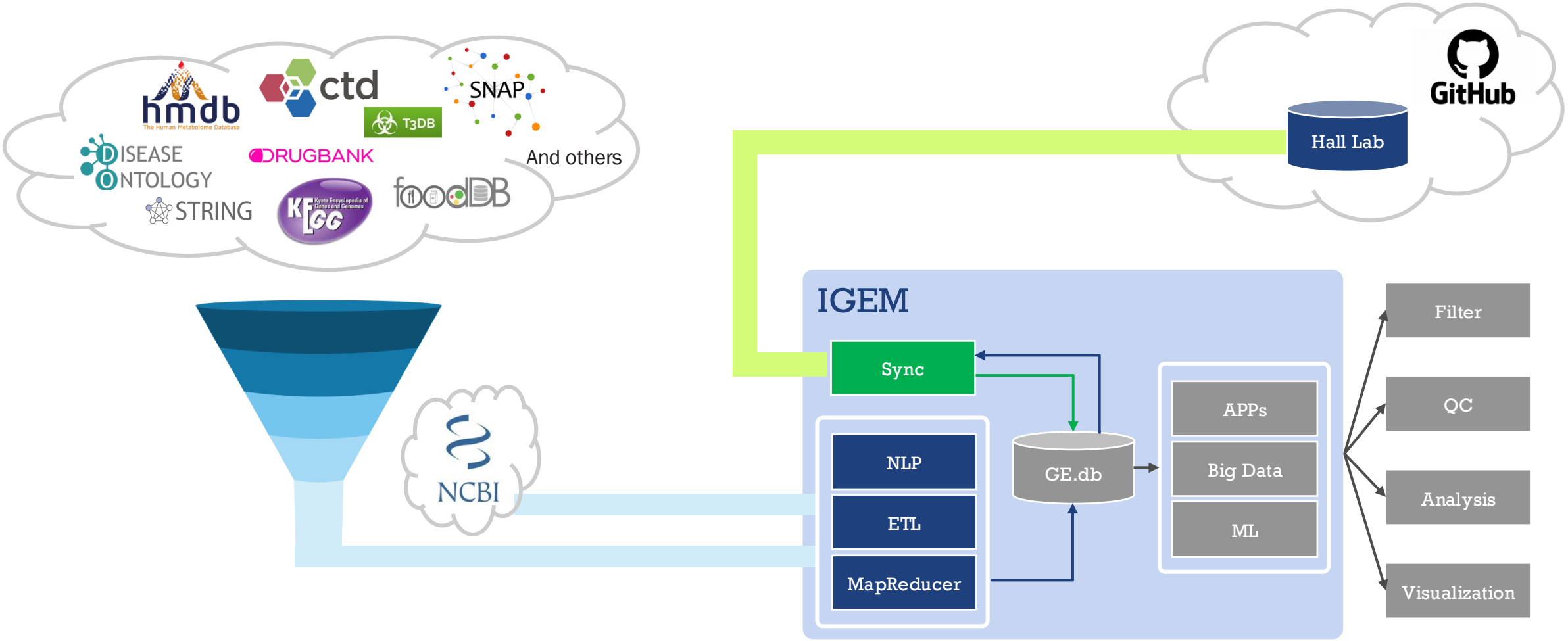
IGEM Overview



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IGEM Overview

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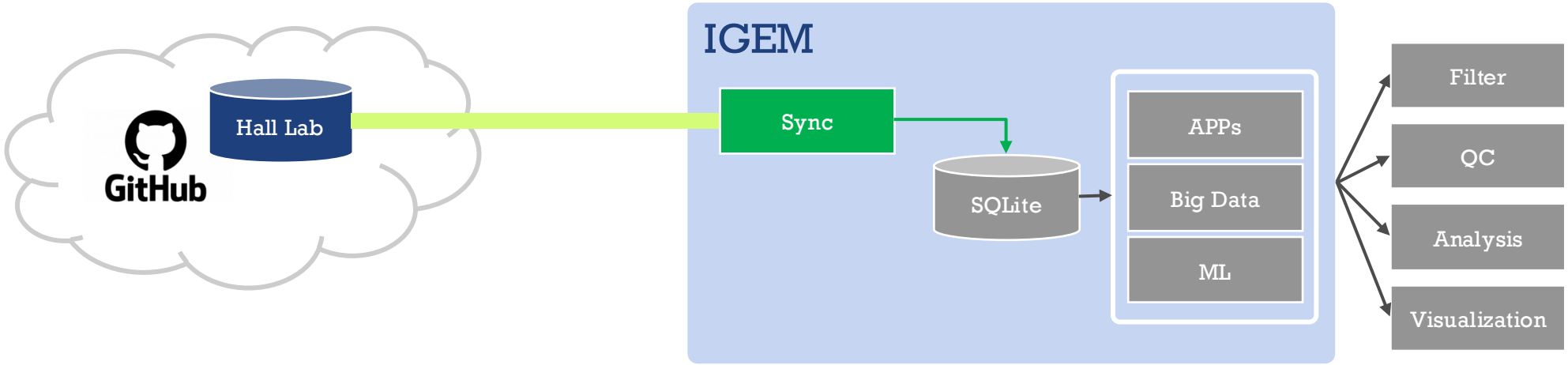
Proposed to work in the ECHO environment

Deploy Client Version

```
Pyhton > 3.8
$ pip install IGEM
On IGEM folder:
    $ python manage.py makemigrations
    $ python manage.py migrate
    if offline:
        Copy dbs files from GitHub to igem/psa/igem
        $ python manage.py sync –local
    if online:
        $ python manage.py sync --remote
```



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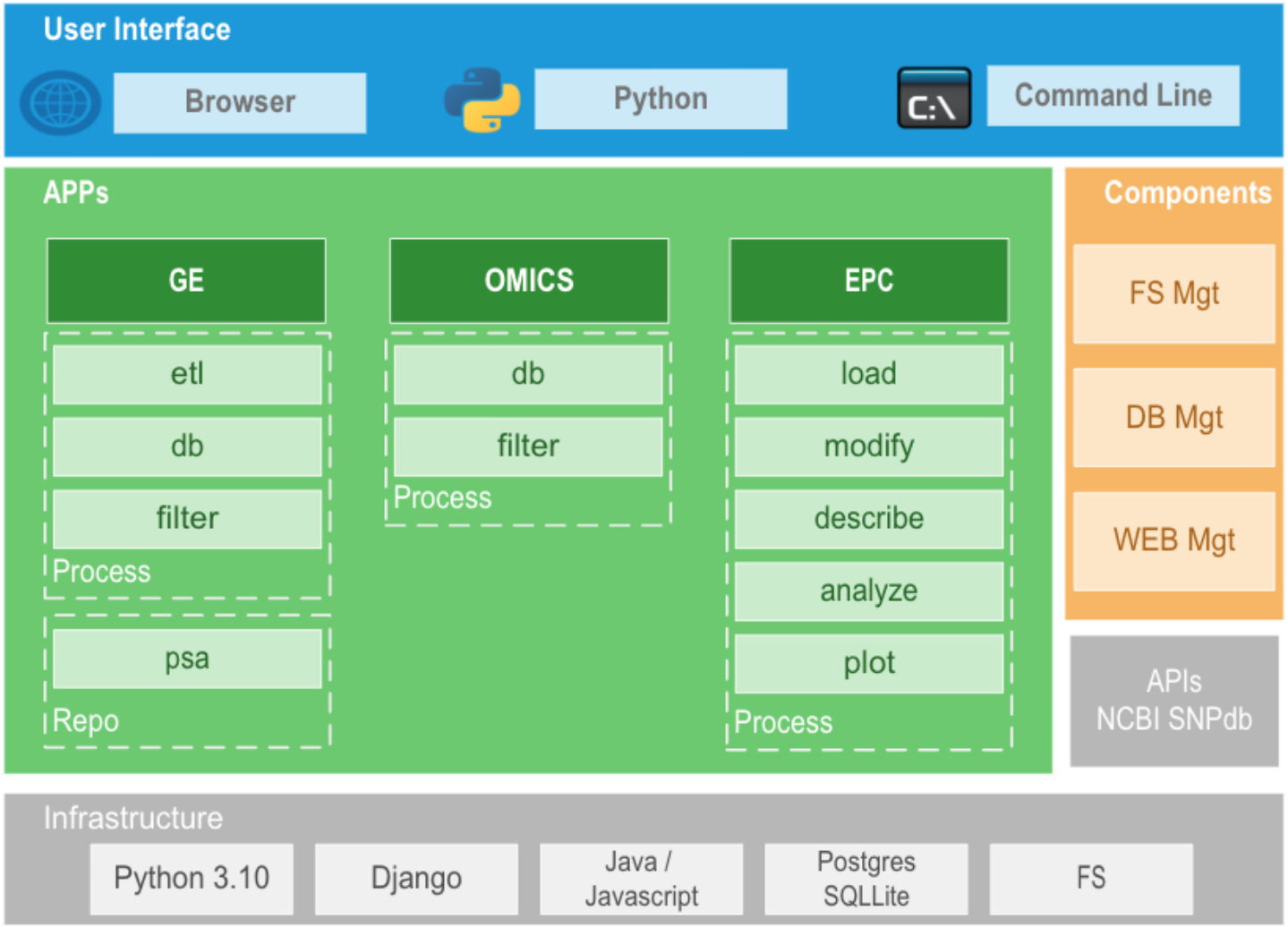
IGEM - Architecture



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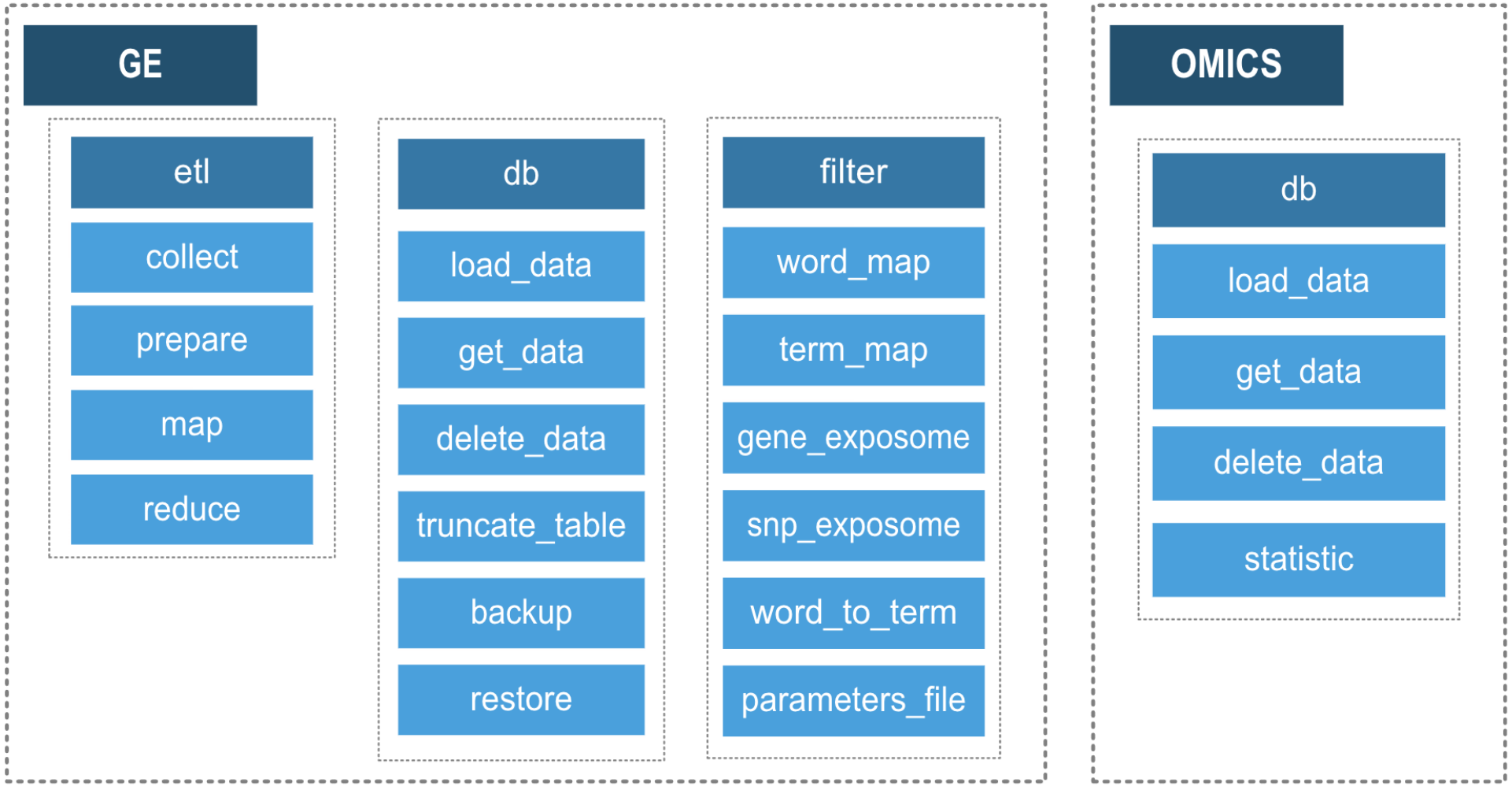
IGEM - Functions



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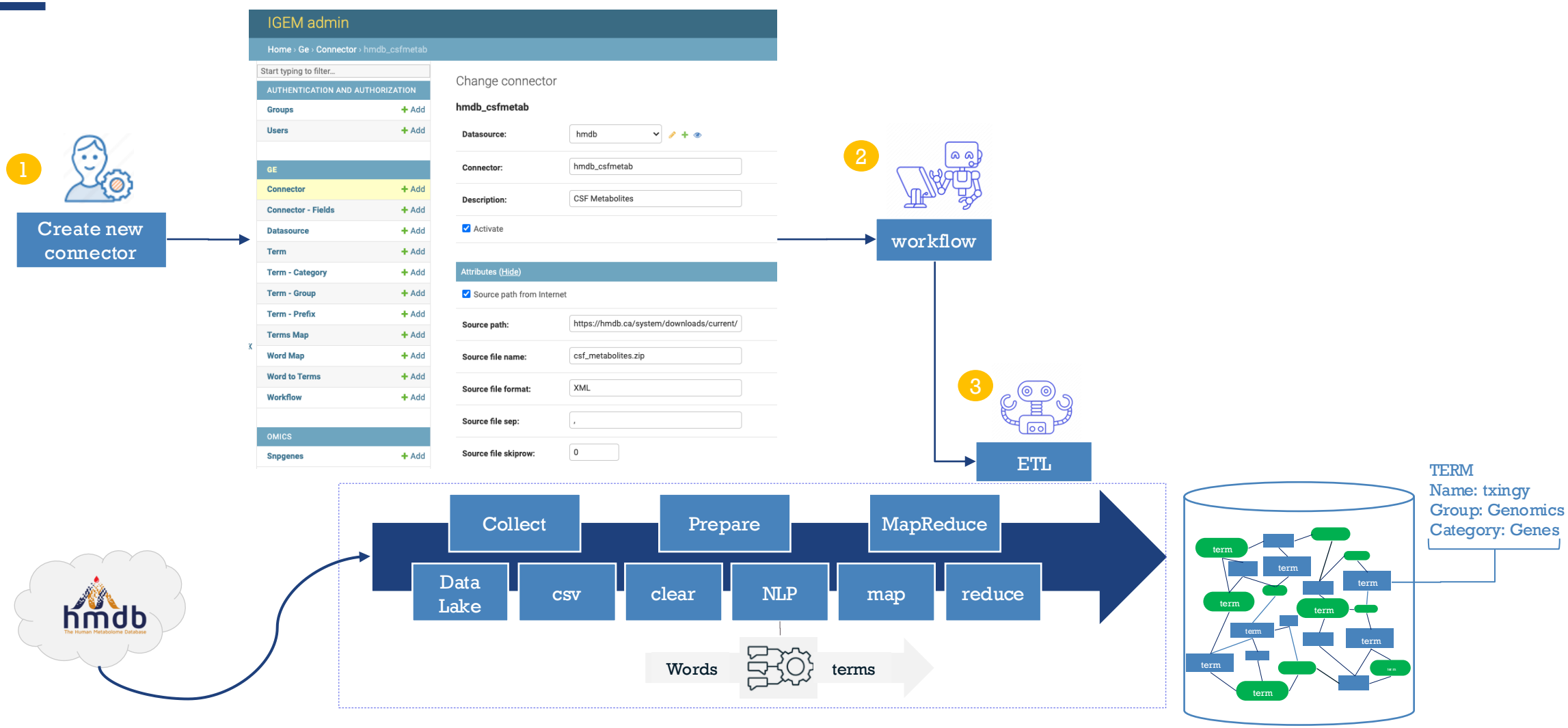
IGEM - Extraction, Processing and Loading



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IGEM - Filter

By Python Script

```
In [2]: from ge import filter

In [3]: df = filter.gene_exposome(term=["gene:246126"])
```

igem > gen246126.csv

	Gene_id	Gene_desc	Exp_grp	Exp_cat	Exp_id	Exp_desc	Qtd_links	Sources	Connector
2	gene:246126	taxilin gamma	environment	chemical	chem:c008261	lead acetate	1	1	ctdcgoasso
0	gene:246126	taxilin gamma	environment	chemical	chem:c511621	2-methyl-2h-pyrazole-3-carboxylic a	2	1	ctdcgint
1	gene:246126	taxilin gamma	environment	chemical	chem:d015124	8-bromo cyclic adenosine monophos	1	1	ctdcgint
*									

By Command Line

```
→ python manage.py filter --gene_exposome 'term=["gene:246126"]'
Run gene_exposome report
  Informed parameters: term=["gene:246126"]

       gene_id                          gene_desc      exp_grp  ... qtd_links sources   connector
2  gene:246126  taxilin gamma pseudogene, y-linked  environment  ...         1       1  ctdcgoasso
0  gene:246126  taxilin gamma pseudogene, y-linked  environment  ...         2       1    ctdcgint
1  gene:246126  taxilin gamma pseudogene, y-linked  environment  ...         1       1    ctdcgint

[3 rows x 9 columns]
```

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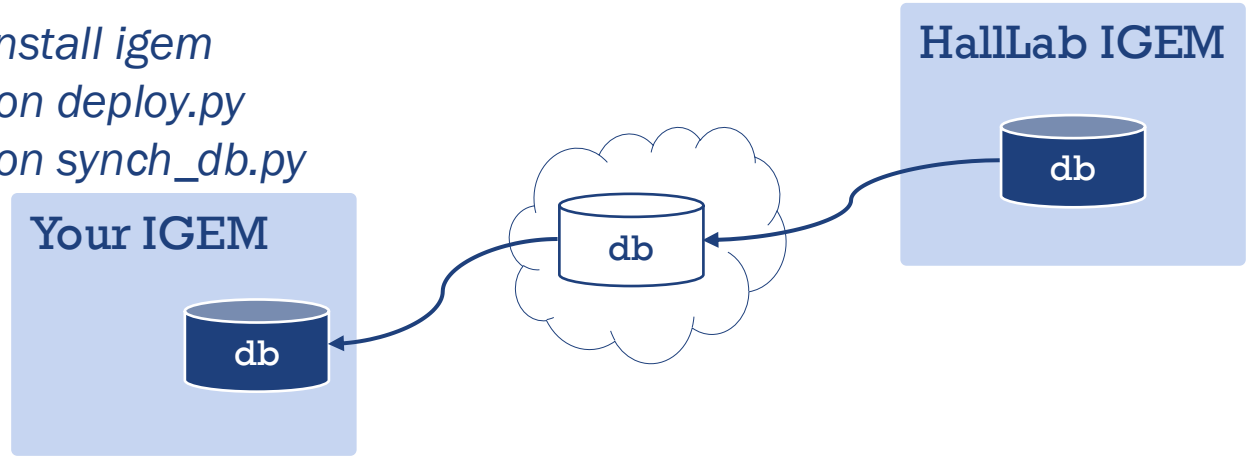
IGEM - Install

Question: Would you like to use HallLab's knowledge base and synchronize new releases,
or would you prefer to manage the external connections and the whole ETL process and create your knowledge base?

$ pip install igem
$ python deploy.py
$ python synch_db.py



$ pip install igem
$ python manage.py makemigrations
$ python manage.py migrate

IGEM - Next developments

- Development of GxE and ExE Interaction Analysis
- KEGG Pathways Base and use in the ETL process
- Implement new WEB interfaces for Filters and Views
- Develop IGEM as Web Services

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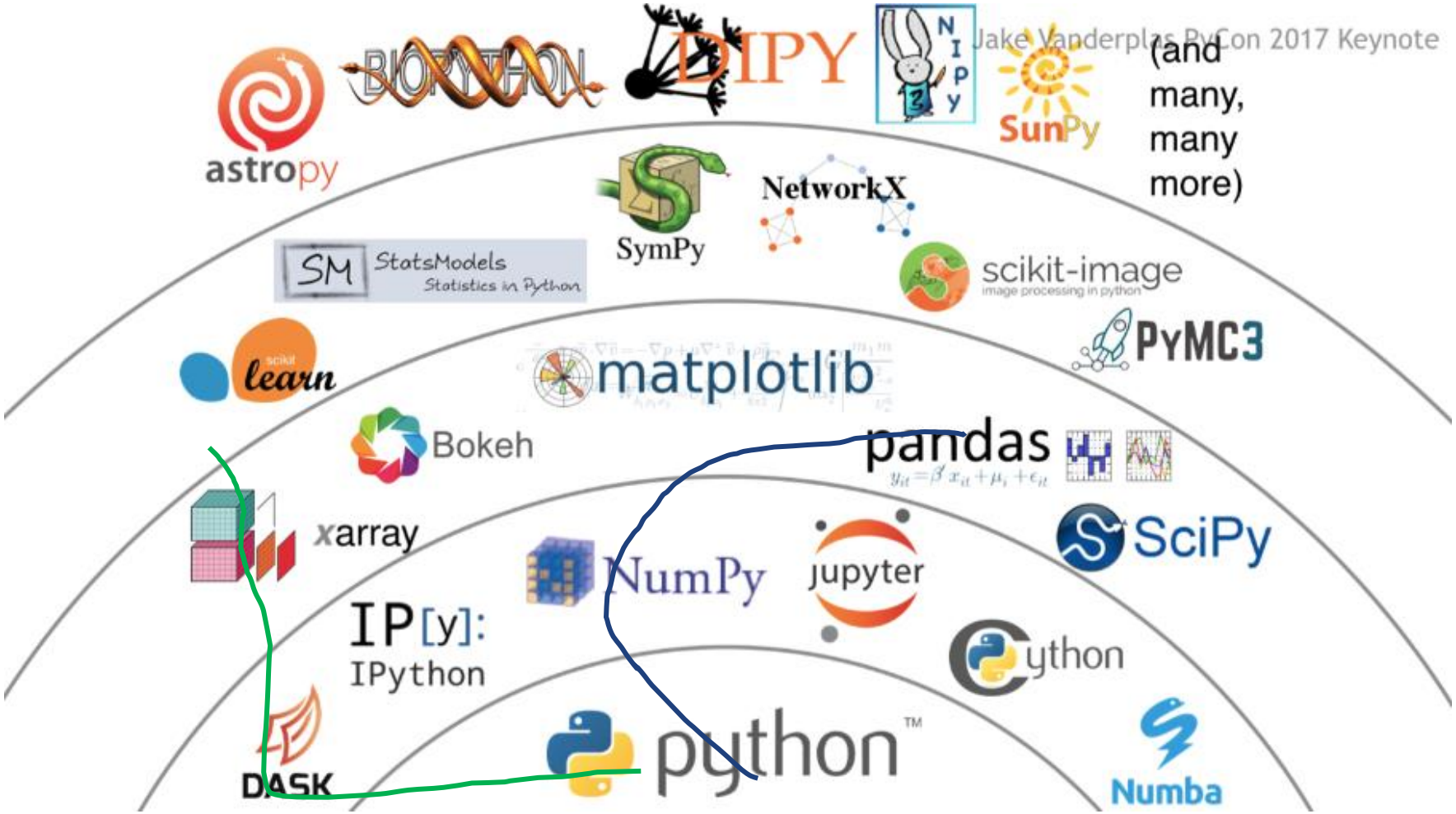
IGEM – Read PLINK



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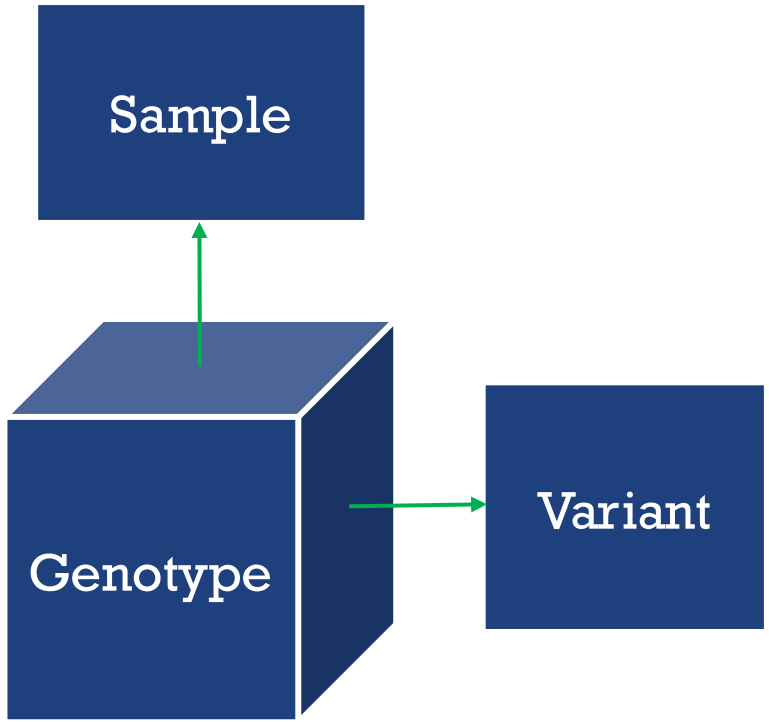
IGEM – Read PLINK




```
import igem

G = igem.load.plink.plink1_xa(bed, bim, fam, verbose=False)
G

<xarray.DataArray 'genotype' (sample: 3061, variant: 687262)>
dask.array<transpose, shape=(3061, 687262), dtype=float32, chunksize=(1024, 1024), chunktype=numpy.ndarray>
Coordinates: (12/14)
  * sample   (sample) object '1' '2' '3' '4' '6' ... '3410' '3412' '3471' '3487'
  * variant  (variant) <U13 'variant0' 'variant1' ... 'variant687261'
    fid      (sample) object '0' '0' '0' '0' '0' '0' ... '0' '0' '0' '0' '0' '0'
    iid      (sample) object '1' '2' '3' '4' '6' ... '3410' '3412' '3471' '3487'
    father   (sample) object '0' '0' '0' '0' '0' '0' ... '0' '0' '0' '0' '0' '0'
    mother   (sample) object '0' '0' '0' '0' '0' '0' ... '0' '0' '0' '0' '0' '0'
    ...       ...
    chrom    (variant) object '1' '1' '1' '1' '1' ... '22' '22' '22' '22' '22'
    snp      (variant) object 'rs12565286' 'rs3094315' ... 'rs28729663'
    cm       (variant) float64 0.0 0.0 0.0 0.0 0.0 0.0 ... 0.0 0.0 0.0 0.0 0.0
    pos      (variant) int32 711153 742429 751595 ... 49522492 49533094 49565872
    a0       (variant) object 'C' 'G' 'C' 'A' 'T' 'A' ... 'T' 'A' 'G' 'T' 'A'
    a1       (variant) object 'G' 'A' 'T' 'T' 'C' 'G' ... 'C' 'G' 'A' 'C' 'G'
```

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IGEM – Read PLINK

```
import igem

G = igem.load.plink.plink1_xa(bed, bim, fam, verbose=False)
G

<xarray.DataArray 'genotype' (sample: 3061, variant: 687262)>
dask.array<transpose, shape=(3061, 687262), dtype=float32, chunksize=(1024, 1024), chunktype=numpy.ndarray>
Coordinates: (12/14)
  * sample   (sample) object '1' '2' '3' '4' '6' ... '3410' '3412' '3471' '3487'
  * variant  (variant) <U13 'variant0' 'variant1' ... 'variant687261'
    fid      (sample) object '0' '0' '0' '0' '0' '0' ... '0' '0' '0' '0' '0' '0'
    iid      (sample) object '1' '2' '3' '4' '6' ... '3410' '3412' '3471' '3487'
    father   (sample) object '0' '0' '0' '0' '0' '0' ... '0' '0' '0' '0' '0' '0'
    mother   (sample) object '0' '0' '0' '0' '0' '0' ... '0' '0' '0' '0' '0' '0'
    ...       ...
    chrom    (variant) object '1' '1' '1' '1' '1' ... '22' '22' '22' '22' '22'
    snp      (variant) object 'rs12565286' 'rs3094315' ... 'rs28729663'
    cm       (variant) float64 0.0 0.0 0.0 0.0 0.0 0.0 ... 0.0 0.0 0.0 0.0 0.0
    pos      (variant) int32 711153 742429 751595 ... 49522492 49533094 49565872
    a0       (variant) object 'C' 'G' 'C' 'A' 'T' 'A' ... 'T' 'A' 'G' 'T' 'A'
    a1       (variant) object 'G' 'A' 'T' 'T' 'C' 'G' ... 'C' 'G' 'A' 'C' 'G'
```

```
gen_rs12565286 = G.where(G.snp == 'rs12565286', drop=True).values

gen_rs12565286
array([[ 2.],
       [ 2.],
       [ 2.],
       ...,
       [ 2.],
       [nan],
       [ 2.]], dtype=float32)
```

```
c = G.where(G.snp == 'rs12565286', drop=True)
c

<xarray.DataArray 'genotype' (sample: 3061, variant: 1)>
dask.array<where, shape=(3061, 1), dtype=float32, chunksize=(1024, 1), chunktype=numpy.ndarray>
Coordinates: (12/14)
  * sample   (sample) object '1' '2' '3' '4' '6' ... '3410' '3412' '3471' '3487'
  * variant  (variant) <U13 'variant0'
    fid      (sample) object '0' '0' '0' '0' '0' '0' ... '0' '0' '0' '0' '0' '0'
    iid      (sample) object '1' '2' '3' '4' '6' ... '3410' '3412' '3471' '3487'
    father   (sample) object '0' '0' '0' '0' '0' '0' ... '0' '0' '0' '0' '0' '0'
    mother   (sample) object '0' '0' '0' '0' '0' '0' ... '0' '0' '0' '0' '0' '0'
    ...       ...
    chrom    (variant) object '1'
    snp      (variant) object 'rs12565286'
    cm       (variant) float64 0.0
    pos      (variant) int32 711153
    a0       (variant) object 'C'
    a1       (variant) object 'G'
```

```
G.snp.values
pydevd warning: Getting attribute DataArray.values was slow (took 25.18s)
Customize report timeout by setting the `PYDEVD_WARN_SLOW_RESOLVE_TIMEOUT` environment variable to a higher timeout (default is: 0.5s)
array(['rs12565286', 'rs3094315', 'rs2286139', ..., 'rs3810648',
       'rs3865766', 'rs28729663'], dtype=object)
```

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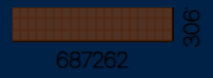
IGEM – Read PLINK

```
G
[14]  ✓ 0.1s

xarray.DataArray 'genotype' (sample: 3061, variant: 687262)
```

	Array	Chunk
Bytes	7.84 GiB	4.00 MiB
Shape	(3061, 687262)	(1024, 1024)
Dask graph	2016 chunks in 4706 graph layers	
Data type	float32 numpy.ndarray	



▾ Coordinates:

sample	(sample)	object	'1' '2' '3' ... '3471' '3487'
variant	(variant)	<U13	'variant0' ... 'variant687261'
fid	(sample)	object	'0' '0' '0' '0' ... '0' '0' '0' '0'
iid	(sample)	object	'1' '2' '3' ... '3471' '3487'
father	(sample)	object	'0' '0' '0' '0' ... '0' '0' '0' '0'
mother	(sample)	object	'0' '0' '0' '0' ... '0' '0' '0' '0'
gender	(sample)	object	'1' '1' '1' '2' ... '1' '1' '1' '1'
trait	(sample)	object	'1' '1' '2' '2' ... '1' '1' '1' '1'
chrom	(variant)	object	'1' '1' '1' '1' ... '22' '22' '22'
snp	(variant)	object	'rs12565286' ... 'rs28729663'
cm	(variant)	float64	0.0 0.0 0.0 0.0 ... 0.0 0.0 0.0 0.0
pos	(variant)	int32	711153 742429 ... 49533094 49565872
a0	(variant)	object	'C' 'G' 'C' 'A' ... 'A' 'G' 'T' 'A'
a1	(variant)	object	'G' 'A' 'T' 'T' ... 'G' 'A' 'C' 'G'

▸ Indexes: (2)

▸ Attributes: (0)

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G x E Interaction (1)

Preparing Data

a. Load PLINK data using ***igem.load.plink1_x()*** and convert it into an xarray.DataArray (let's call it G). Ensure that we have an IID column to identify samples.

b. Load exposome data as a DataFrame (let's call it E). Make sure we have an IID column in E as well.

Merging Data

Merge the G DataArray and E DataFrame based on the IID column. We can use pandas' merge function to achieve this. This will create a new DataFrame containing the IID, Genotype, and Exposome data.

Regression Data

a. Iterate through each column in the Genotype (G) DataArray. For each column:
i. Extract the column as a 1D numpy array.
ii. Merge this array with the Exposome DataFrame based on the IID column.
iii. Use **igem.epc.analysis.study_interaction()**
b. Store the regression results in a suitable data structure, such as a dictionary or a DataFrame.

G x E Interaction (2)

the provided code structure will run the interaction analysis in parallel using the `dask.delayed` mechanism. The `dask.delayed` function allows you to create delayed computation objects, which are essentially placeholders for computations that will be executed in parallel when call `dask.compute`.

In the code snippet I provided, the interaction analysis for each combination of variant and exposome is encapsulated within a `dask.delayed` object. These delayed objects are collected in the `interactions` list. When you call `dask.compute(*interactions)`, Dask will execute these computations in parallel, utilizing the available resources such as CPU cores.

The key steps for parallel execution are:

1. Encapsulate the interaction analysis using `dask.delayed`.
2. Create a list of these delayed objects for all interactions.
3. Use `dask.compute` to trigger the parallel execution of the delayed objects.

However, please note that Dask's parallelism mechanism works best for tasks that can be divided into independent computations, which seems to be the case for your interaction analysis. Additionally, proper configuration of the Dask cluster or client settings may be required to optimize performance based on your hardware resources.